



UNIVERSITY OF
WATERLOO

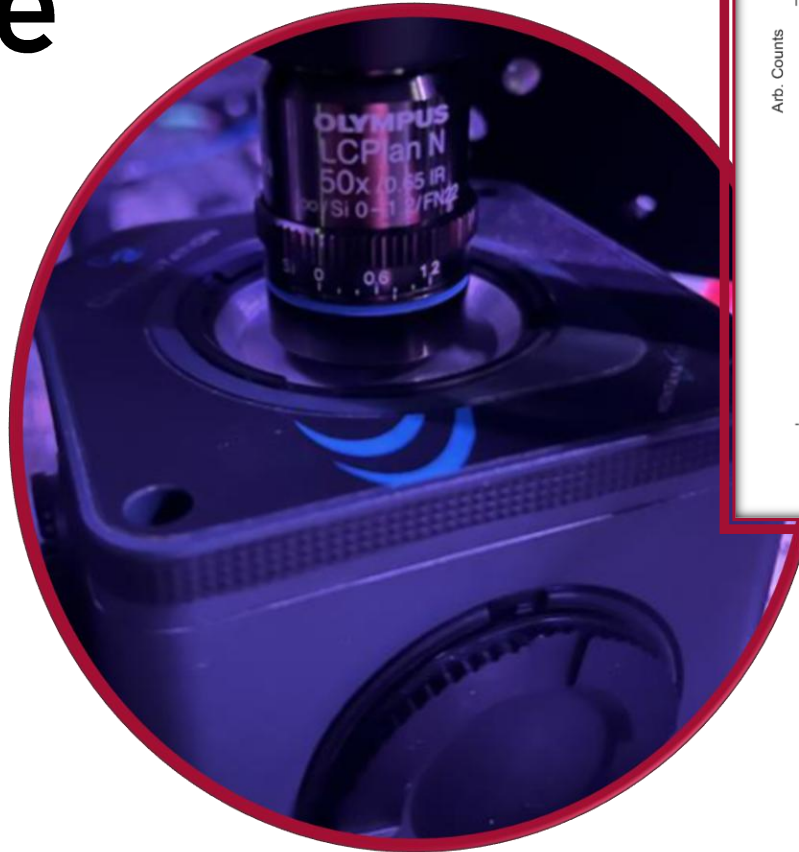
IQC Institute for
Quantum
Computing

Finding the Perfect Quantum Dot (QD)

Bera Emin Yavuz

Undergrad Research Co-op

Cryostat Nanowires Sits in



Angles 0 -180 Degrees (QD [10 4])

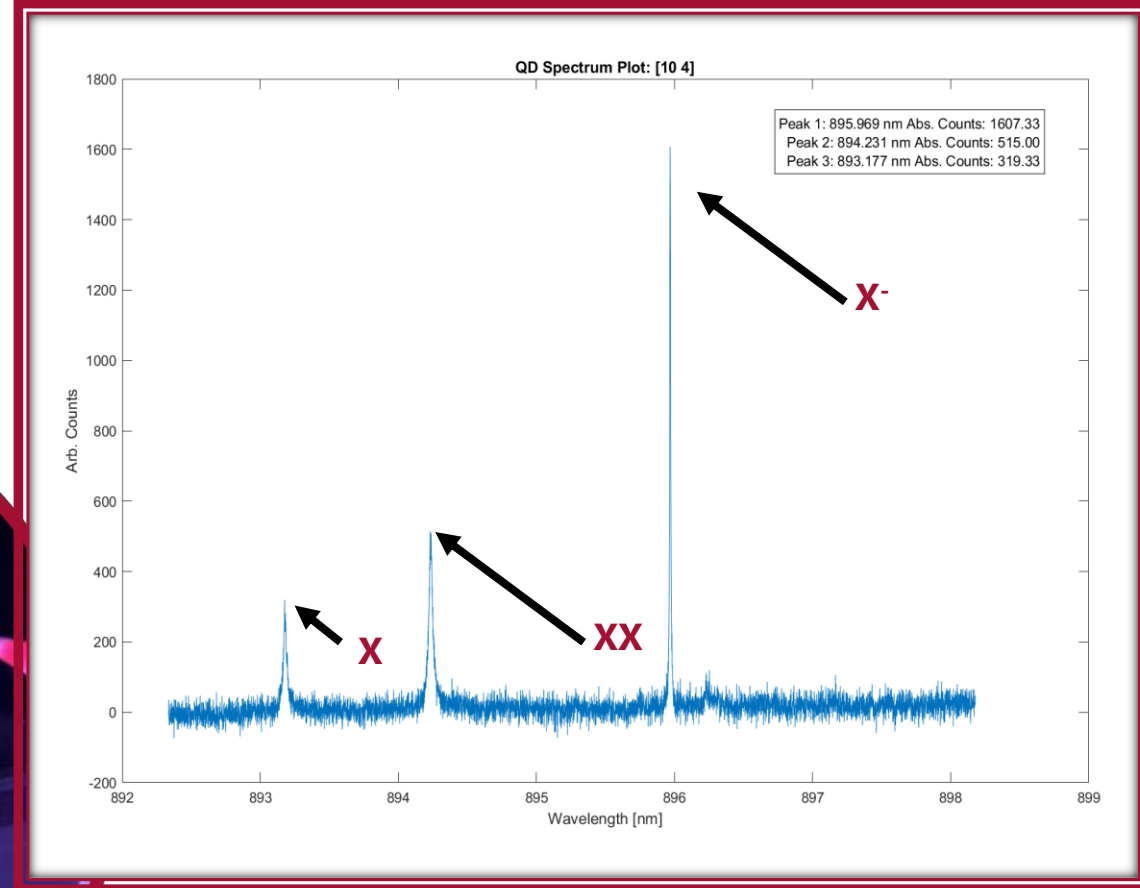


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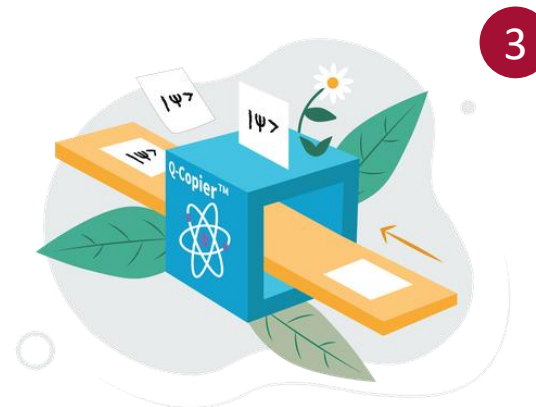
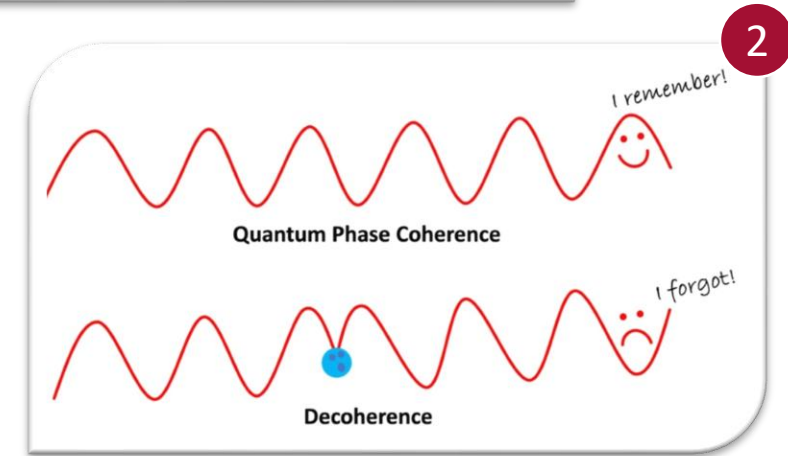
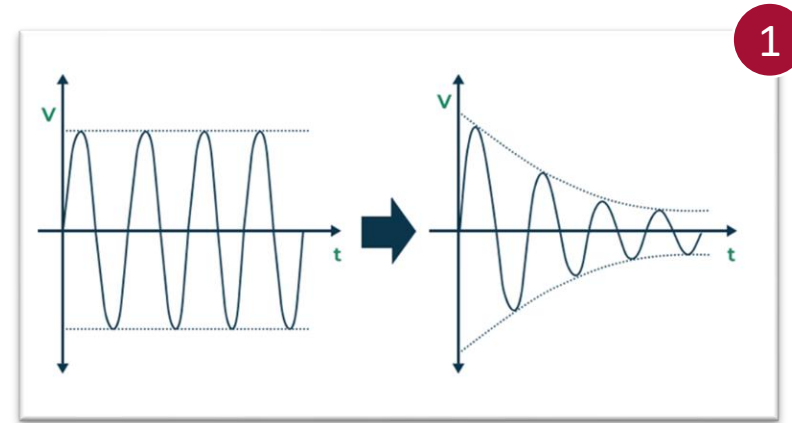
NWQD Operation Application Overview

Side Projects (Fast Photo Diode, 3D Prints)

Big Picture

Quantum Repeater (End Goal)

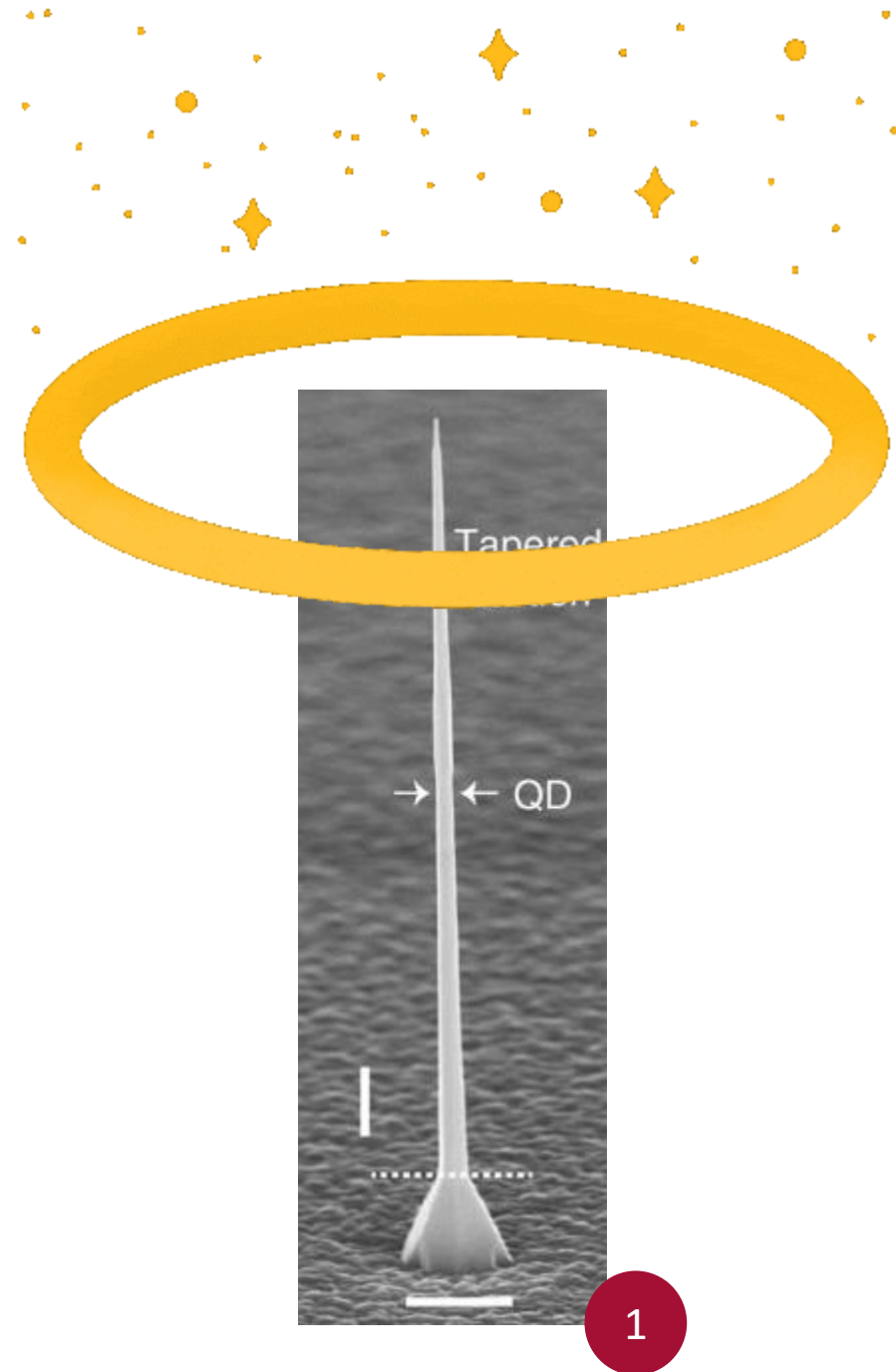
- Crucial component in long-distance quantum communication
- Extends the range of quantum communication by overcoming limitations;
 - **Signal attenuation**
 - **Decoherence**
 - **No-Cloning Theorem**
- Need the perfect entangled photon source **A.K.A find the perfect QD**



What is the Perfect QD?

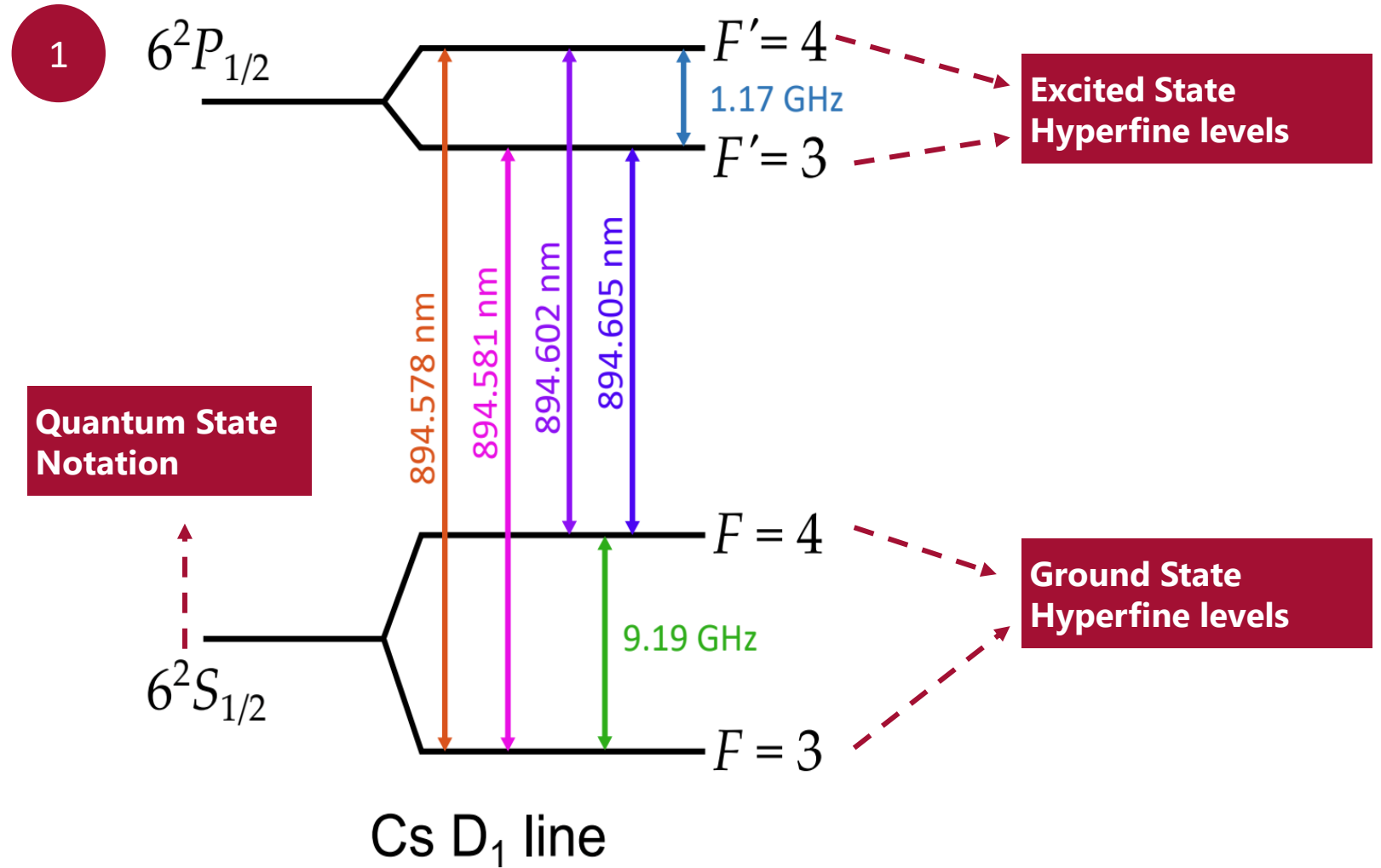
1. QD must "talk" with **Cs atom**
2. Must be **Considerably Bright**
3. Produce Good **Quality Entangled Photons**

**But what are these
things exactly?**

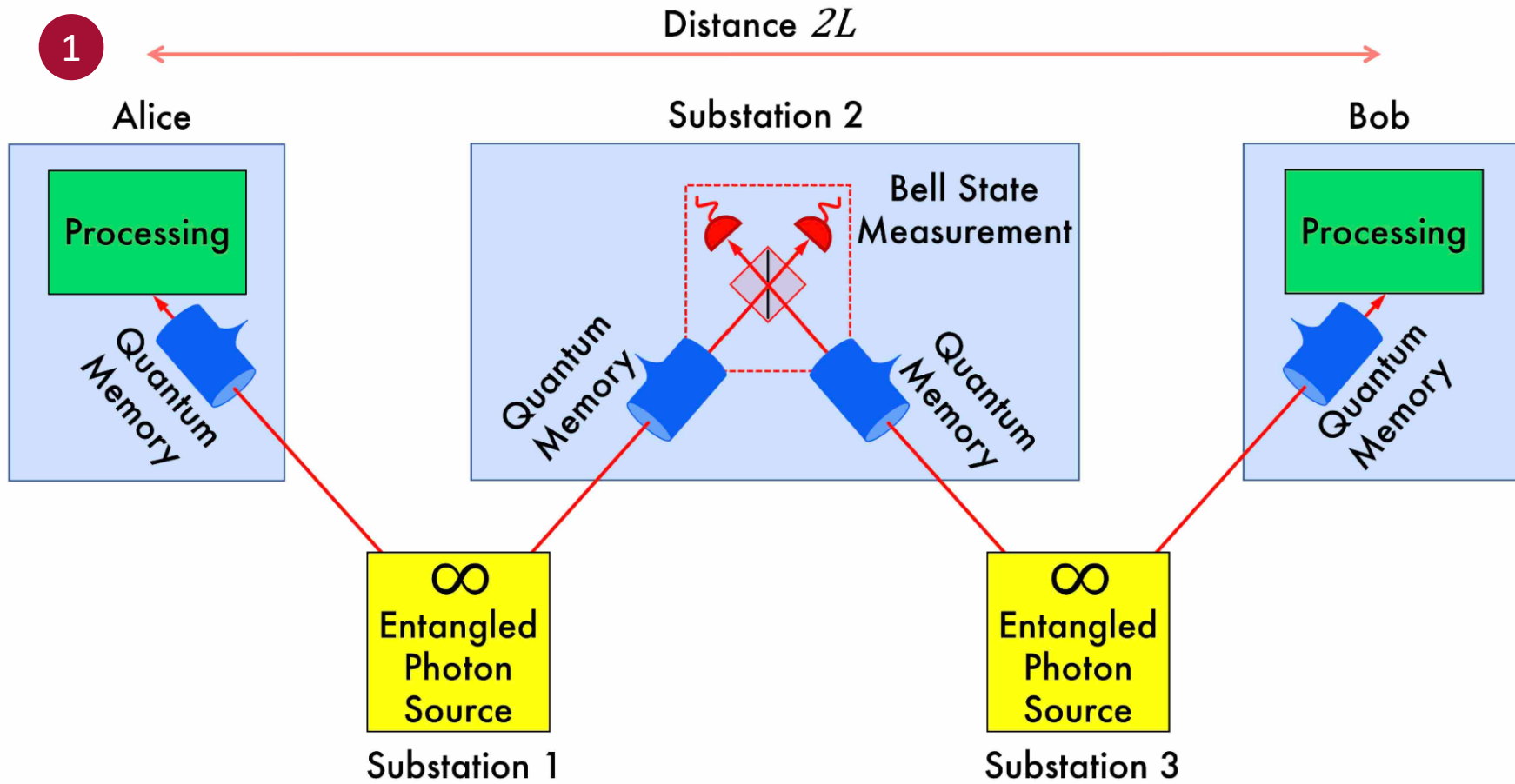


“Talking” with Cesium Atoms

- Photons wavelengths emitted by QD must match the **D1 Transition Line of Cesium**
- **4** Possible “**perfect wavelengths**”
- Matching **transition lines ensures coherent interaction**, improving the **fidelity** of stored quantum states.



Cesium Atom Energy levels (D1 transition line)



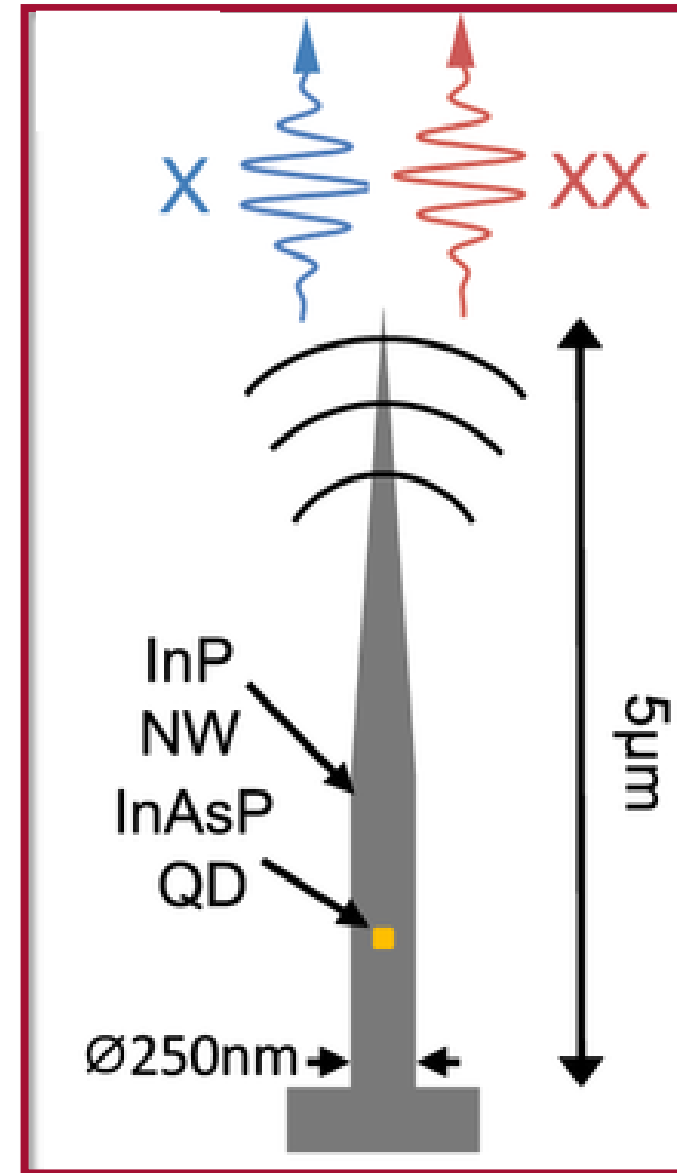
Quantum Memory:

- Caesium (Cs)

Entangled Photon Source:

- Photons emitted from QD

**How do we
excite a QD?**



QD Excitation Simple Overview

Quantum dots = semiconductor nanocrystals

Electrons are **confined** → discrete energy levels

Laser excites an electron → moves to **higher energy level**

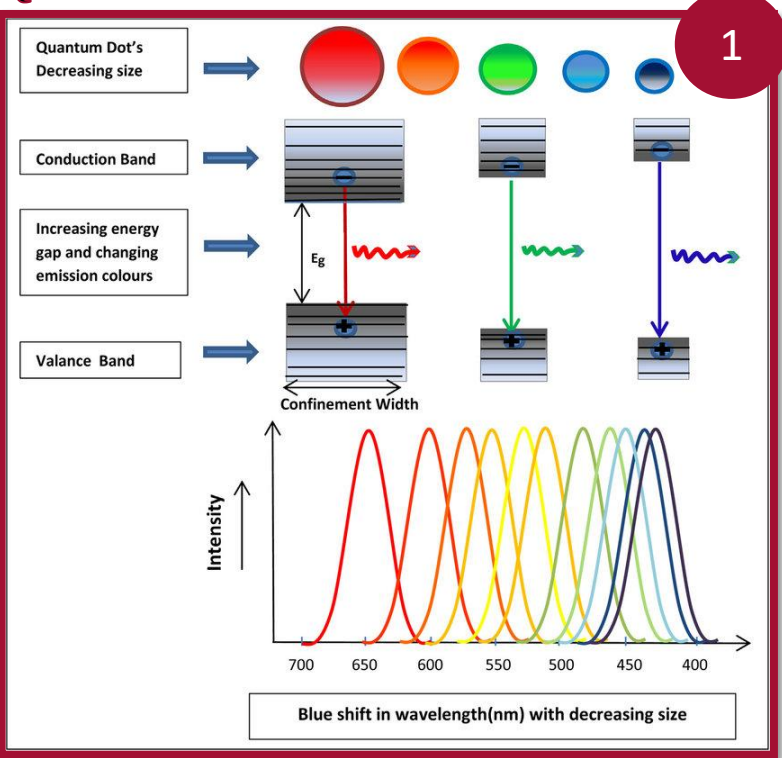
Forms **exciton** (electron + hole pair)

Electron falls back → emits a **photon** (light)

Exciton Formation

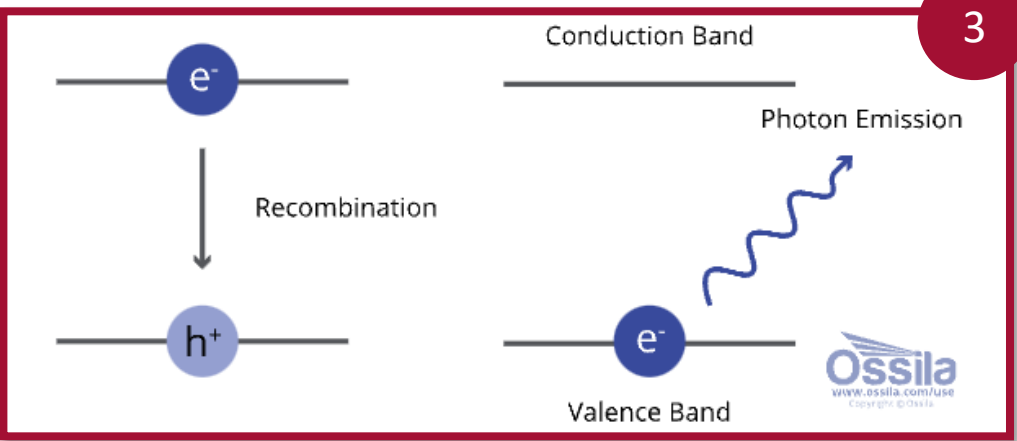


Quantum Confinement Effect



1

Photon Emission



3

Cs atoms

4



Nanowire QD

QD Emitting Photon

1. Google Images
2. Google Images
3. Google Images
4. Fine Structure Splitting of Quantum Dots By Katie McDonnell

Different Methods of Excitation

a) Above-Band

Laser Requirements:

- Any laser **above the bandgap** of the barrier material (e.g., **808 nm** for InP)
- No fine tuning** needed

Pros

- Simple and reliable to implement
- High photon emission rates

Limitations:

- Poor photon coherence
- High background noise and jitter

b) Quasi-resonant

Laser Requirements:

- Precisely tuned** to QD **p-shell transition**
Requires **moderate spectral precision** (within ~ 1 nm)

Pros:

- Better **photon purity** and **indistinguishability** than above-band
- Lower background fluorescence

Limitations:

- Some jitter due to relaxation
- Laser tuning must match QD excitation level

c) Resonant

Laser Requirements:

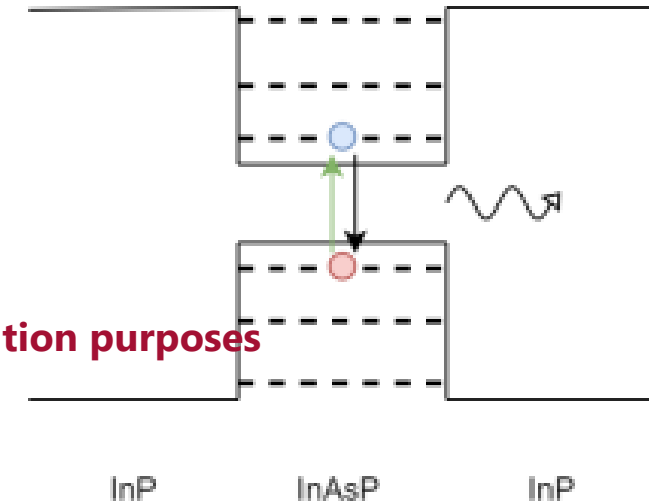
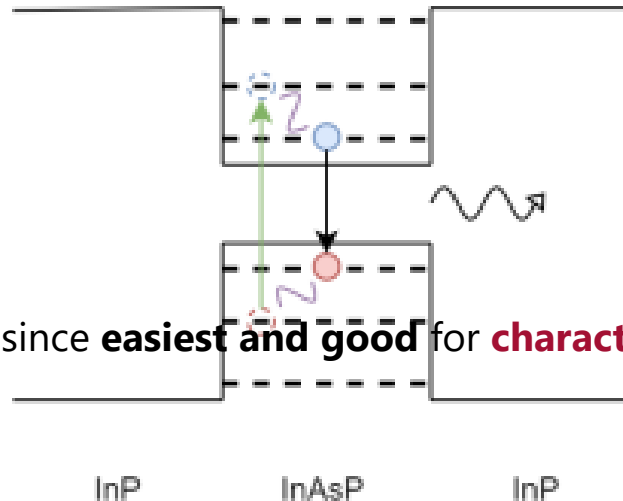
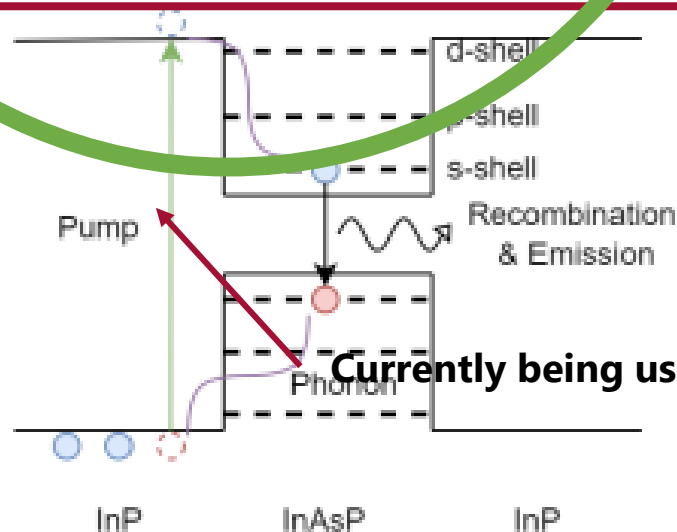
- Precisely tuned** to QD **s-shell transition**
- Requires **narrow-linewidth laser** (precision $\sim$ MHz)

Pros:

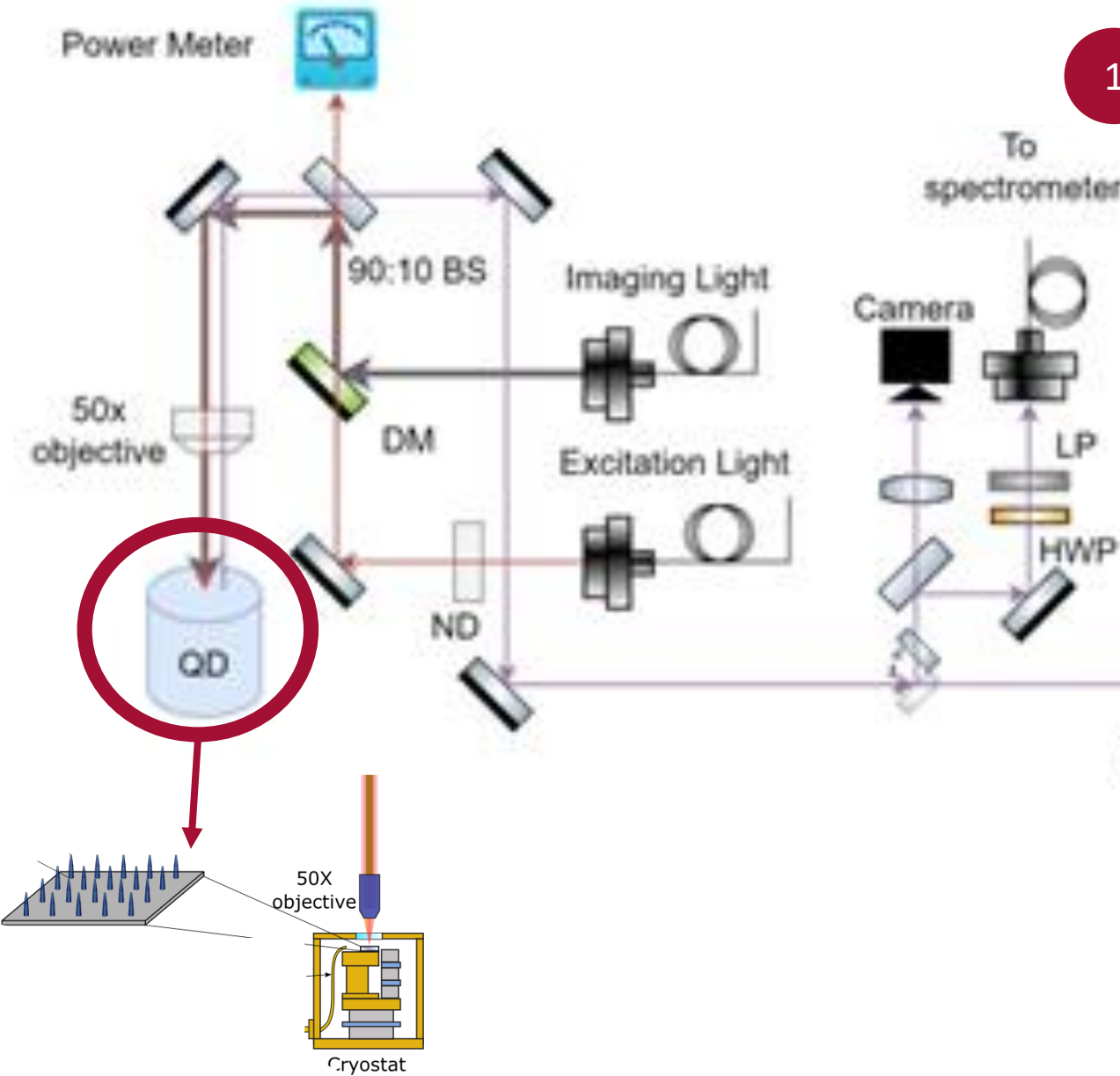
- Best photon coherence** and **indistinguishability**
- Ideal for **quantum interference** and **entanglement protocols**

Limitations:

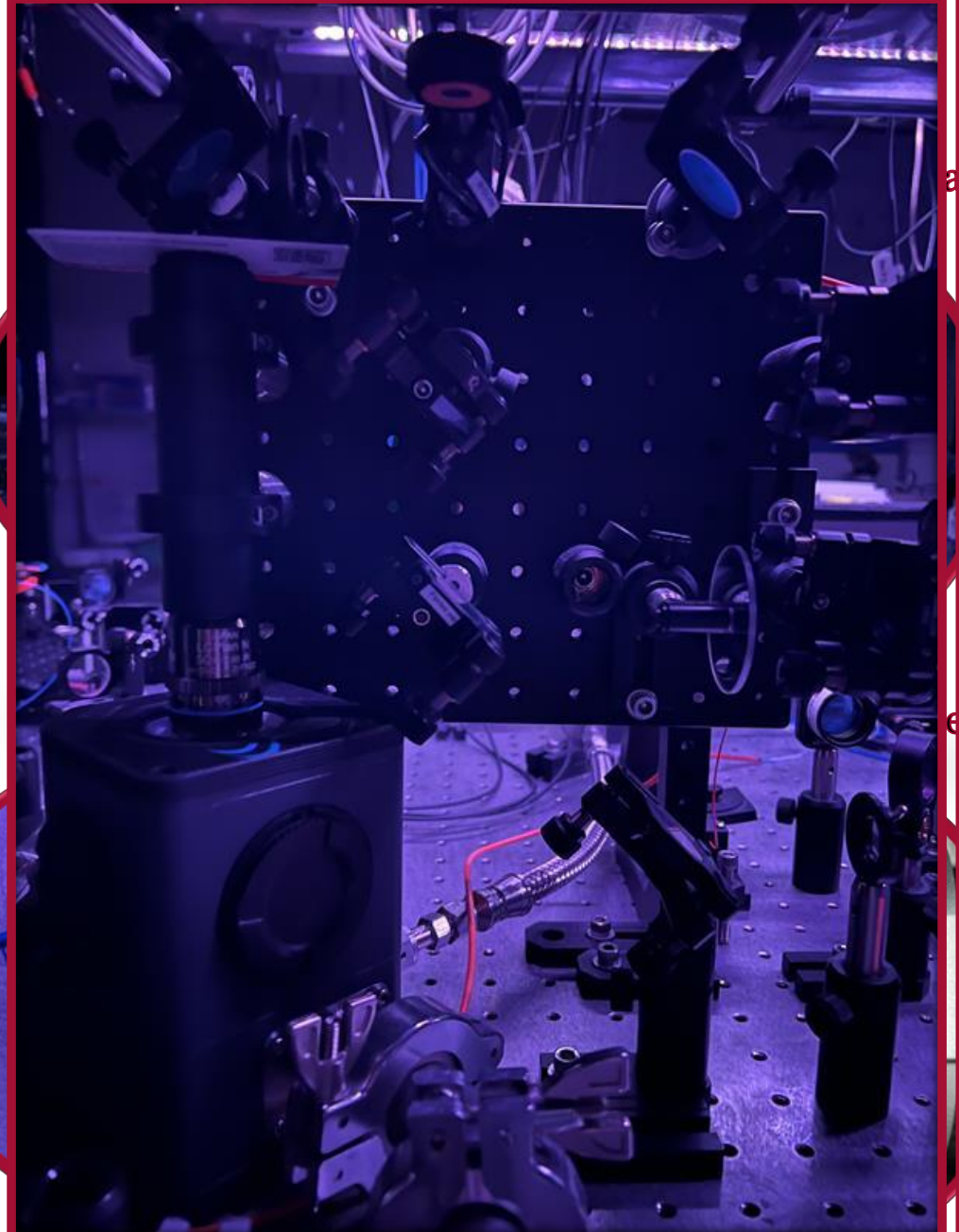
- Technically complex (laser filtering, alignment)
- Requires active stabilization



Optical Setup



1



Spectrometer Resolution Explained

- Spectrometer **relies on diffraction grating** with three different faces;
1800, 1200 and 150 lines/mm face.
 - **Higher groove density (1800 mm)** spreads **out light more**, translating to better separation between close wavelengths
 - **Lower groove density (150 mm)** spreads out light less, meaning wavelengths aren't well-defined

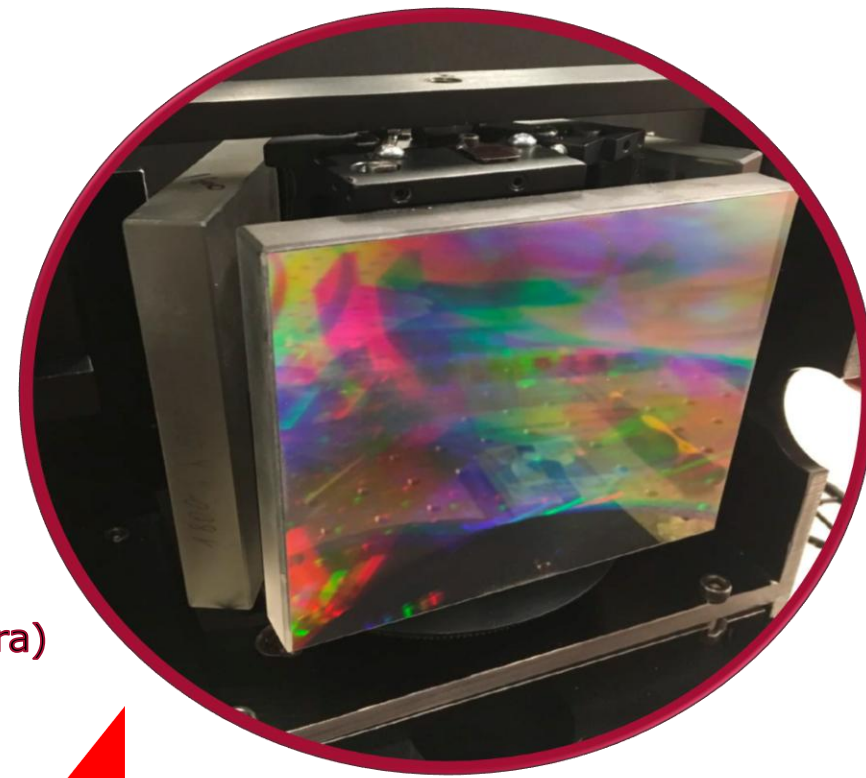
Image Scale

1800 lines/mm: 1pm /pixel

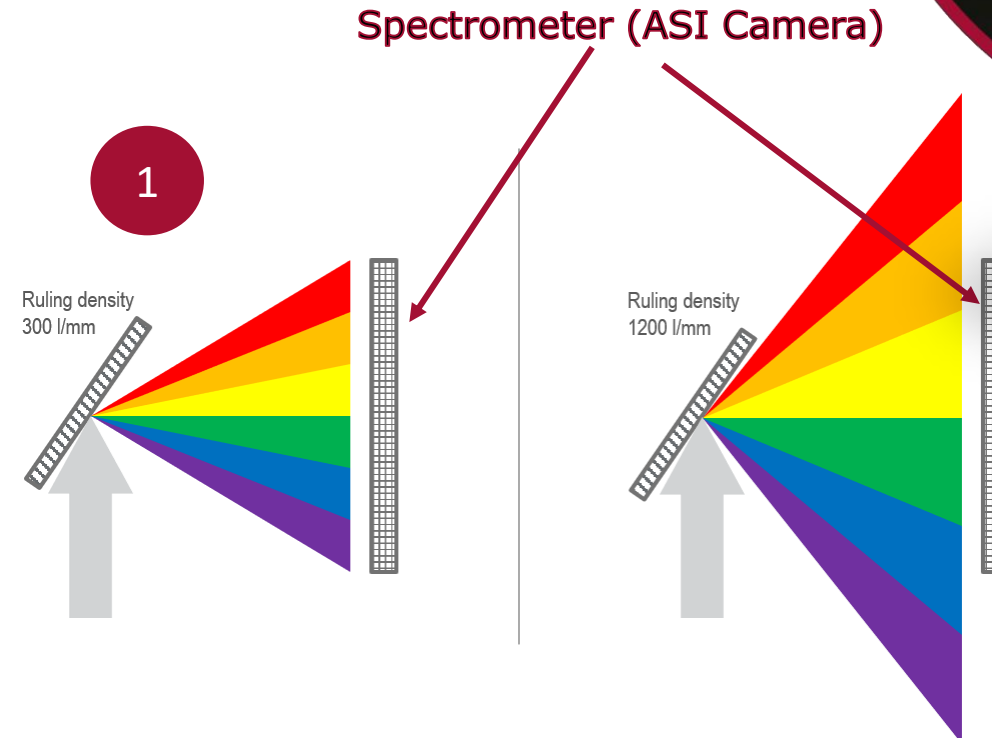
1200 lines/mm: 2 pm/pixel

150 lines/mm: 16 pm/pixel

2



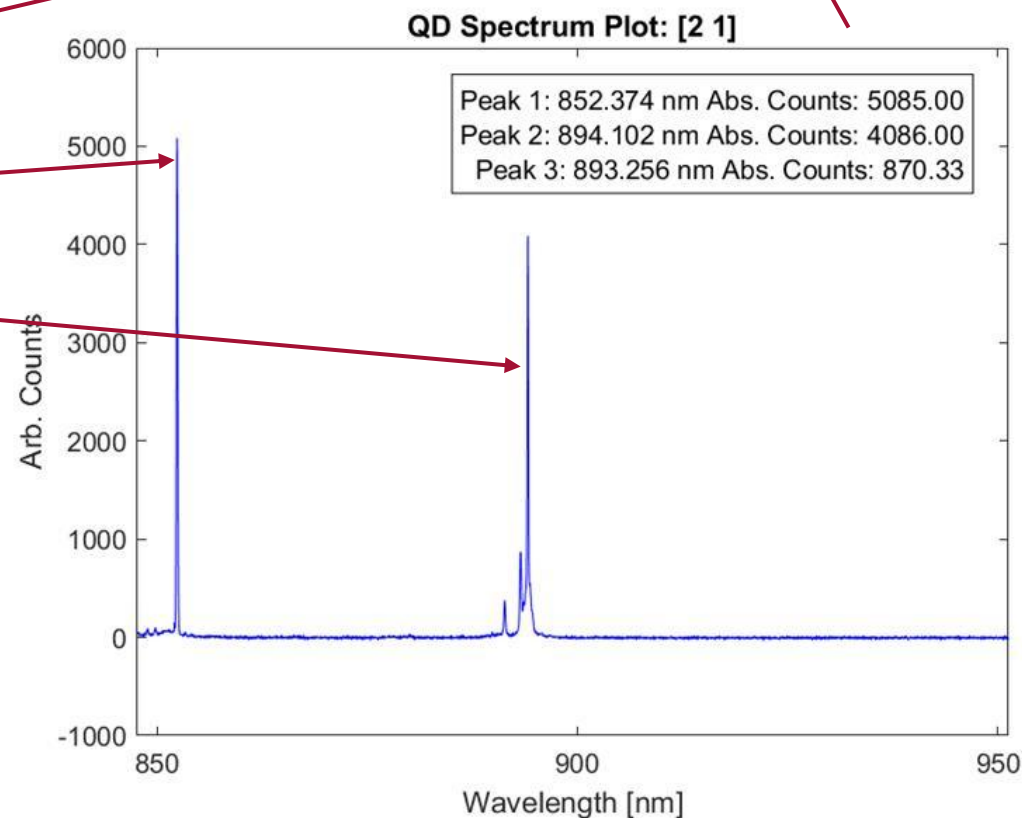
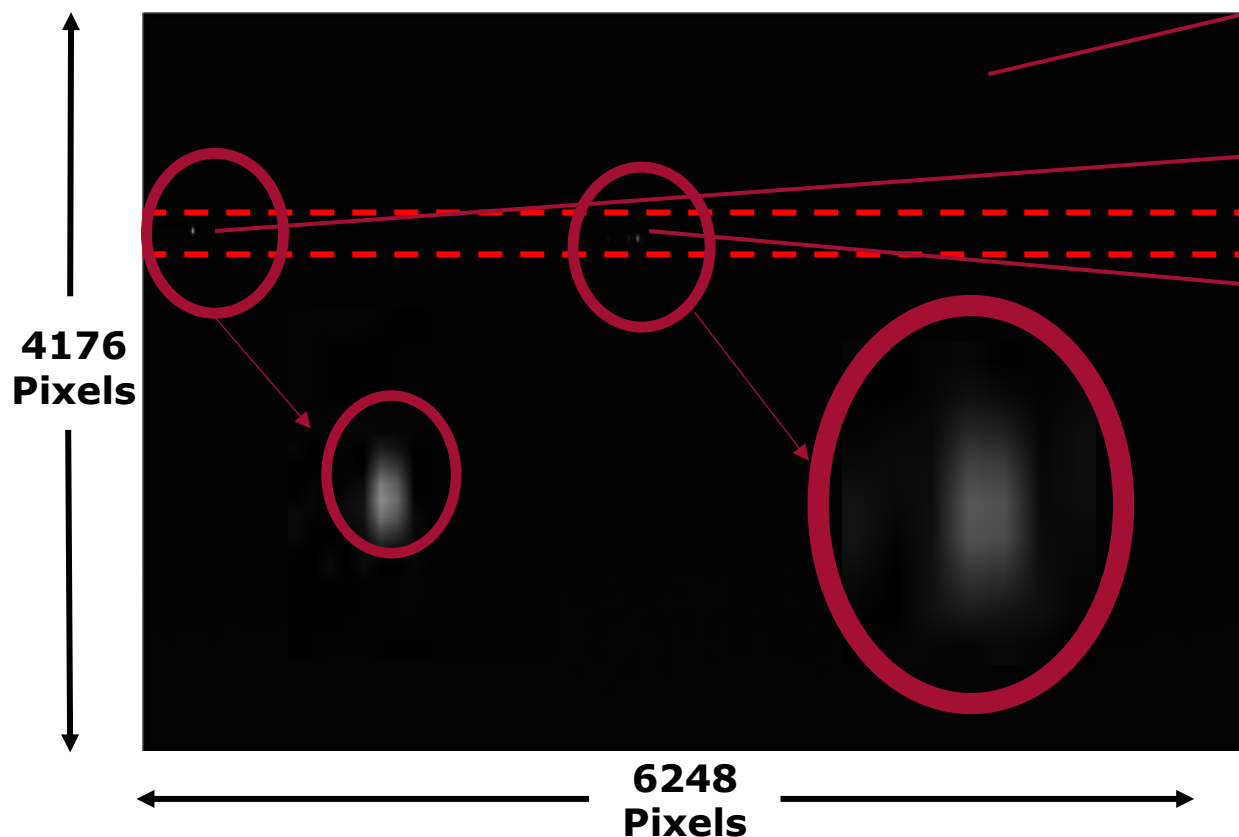
Triple Diffraction Grating



How does the Spectrometer Work (Basics)

- Incoming light is dispersed into its component wavelengths using our diffraction grating.
- The dispersed light forms a two-dimensional spectral image on the camera sensor.
 - **6248 pixels in width by 4176 pixels in height**
- To obtain the spectrum, the pixel values in each column are summed across all rows. This process collapses the **two-dimensional image** into a **one-dimensional spectrum**, displaying **Arb. Counts** versus **wavelength**.

Raw ASI image alongside it's respective **spectrum plot for [2 1]** at the **1200 mm grating**



Raster Scan Overview



Need for Automation

- QDs are grown on a chip (**200x200 grid-like pattern**)
- **40,000 possible QD** emit~895 nm wavelength photons
- **Photons wavelengths:** Standard Deviation of 5nm.
- Use **1200 or 150 line/mm** diffraction grating, better for characterization

Data Saved

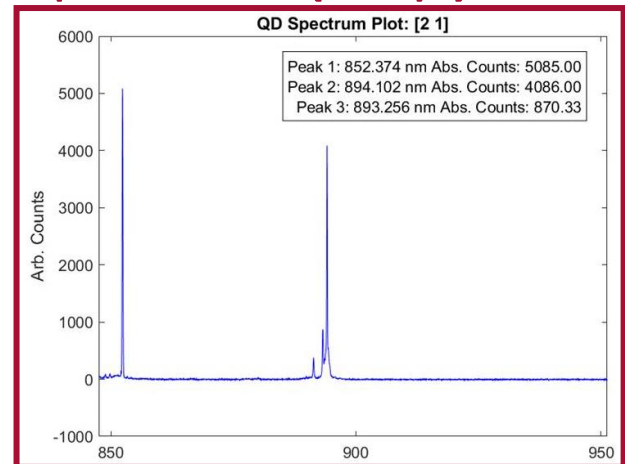
- QD **spectrum plots and text files** saved by **default** with the
- User can **optionally opt** to save **raw QD spectrum** and **white light images**
- **Refer to examples below for QD [2 1]**

UI Tab for Controlling Raster Scan

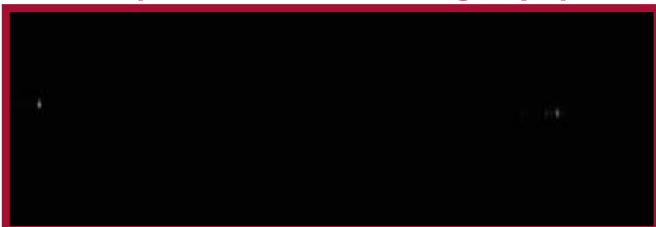
Text File Data (Always)

```
|-----Settings-----  
Spectrometer Grating: 150mm  
Exposure: 1.0000E+00 seconds  
Gain: 300.0  
Resolution: 6248 x 4176  
Captured time: 17:50:41 Mar-17-2025  
-----Data-----  
Wavelen. (nm) Count Average  
847.6614 4.700000e+01  
847.6780 3.500000e+01  
847.6946 5.466667e+01  
847.7112 4.533333e+01  
847.7278 3.433333e+01  
847.7444 3.633333e+01  
847.7609 3.100000e+01  
847.7775 3.166667e+01  
847.7941 2.800000e+01  
847.8107 5.233333e+01  
847.8273 4.433333e+01  
847.8439 2.666667e+01
```

Spectrum Plot (Always)



Raw Spectrometer Image (Optional)



Position Photo (Optional)



Data/Results

- Total data collected: 50 columns
- Data analyzed: 7466 QDs (Columns 14-50)
- 1 Column = 200 unique dots

Data Collection Time Line

AUTOMATED SCANS WITH 150 ln/mm GRATING SUMMARY			
From	To	Date	Comments
[1 1]	[109 6]	06-03-2025	No reference (852 nm) signal ; Raster scan
[118 6]	[153 7]	10-03-2025	No reference (852 nm) signal ; Raster scan
[157 7]	[144 14]	11-03-2025	No reference (852 nm) signal ; Raster scan
[1 2]	[98 4]	21-03-2025	With reference (852 nm) signal; List scan; <i>Note -</i>
[98 4]	[148 14]	24-03-2025	With reference (852 nm) signal; List scan; <i>Note -</i>
[140 14]	[146 22]	25-03-2025	With reference (852 nm) signal; Raster scan
[145 22]	[24 28]	26-03-2025	With reference (852 nm) signal; Raster scan
[24 28]	[1 37]	27-03-2025	With reference (852 nm) signal; Raster scan
[1 37]	[31 39]	28-03-2025	With reference (852 nm) signal; Raster scan
[1 39]	[84 43]	01-04-2025	With reference (852 nm) signal; Raster scan
[85 43]	[1 50]	02-04-2025	With reference (852 nm) signal; Raster scan
Finished 50 columns (10,000) QD: 1/4 Mark			

9:44 hours

7:30 hours

9:47 hours

2:32 hours

5:34 hours

8:00 hours

→

→

→

→

→

→

←

←

←

←

←

←

~ 8 columns done

~ 6.5 columns done

~ 8 columns done

~ 2 columns done

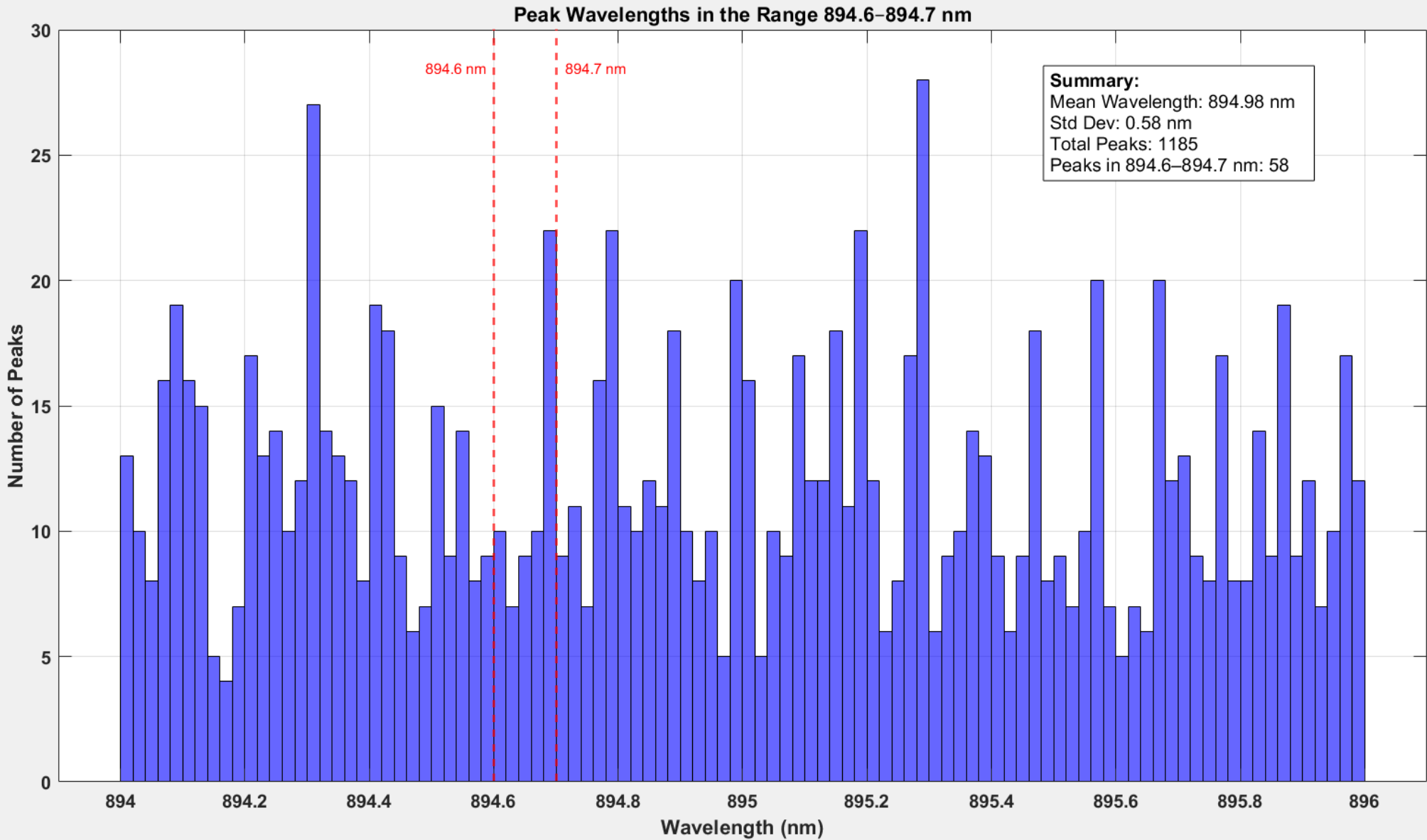
~ 4.25 columns done

~ 7.5 columns done

Data/Results

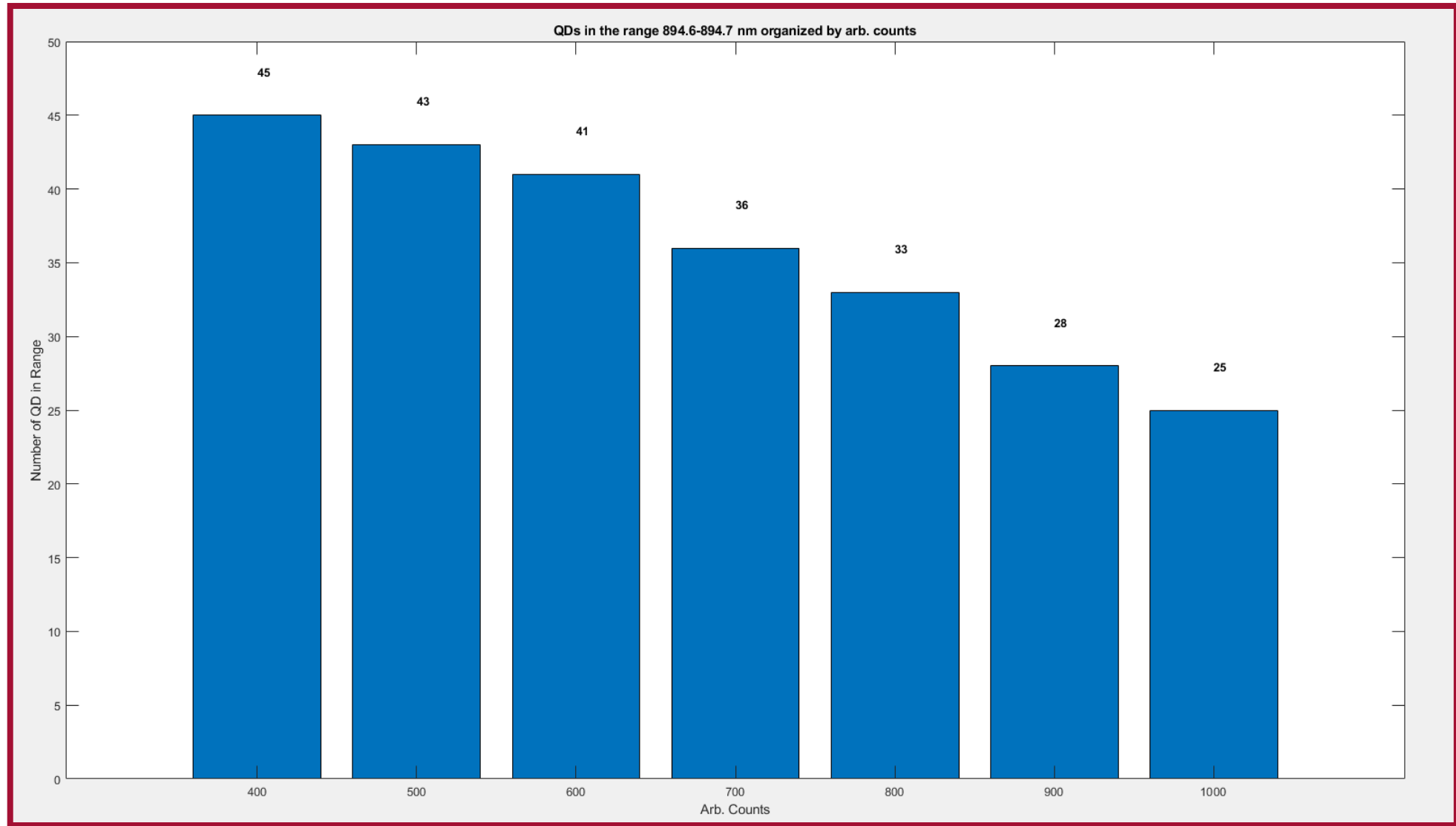
This Histogram has a peak cut off of 300 Counts

Number of Peaks

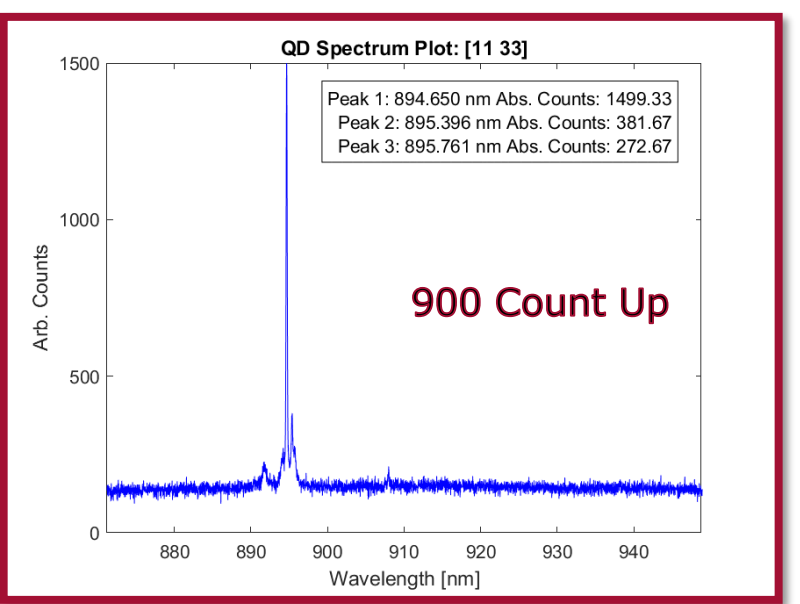
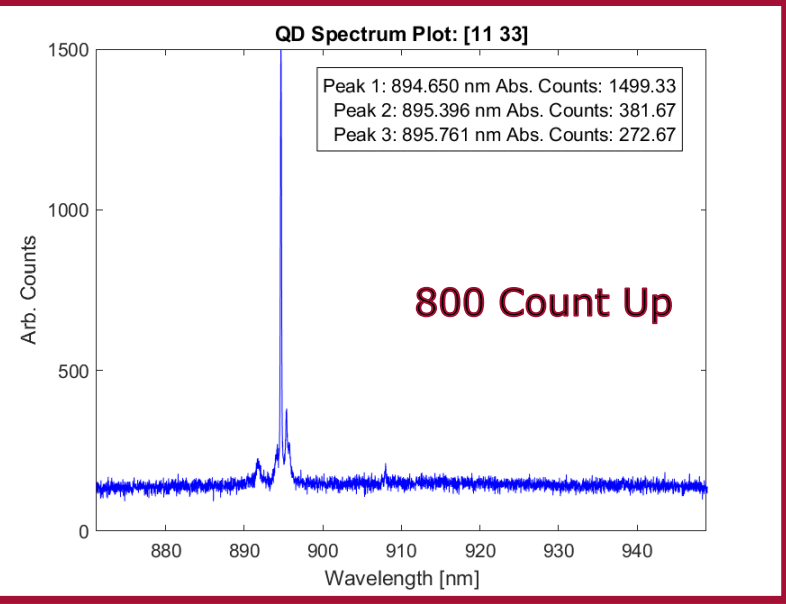
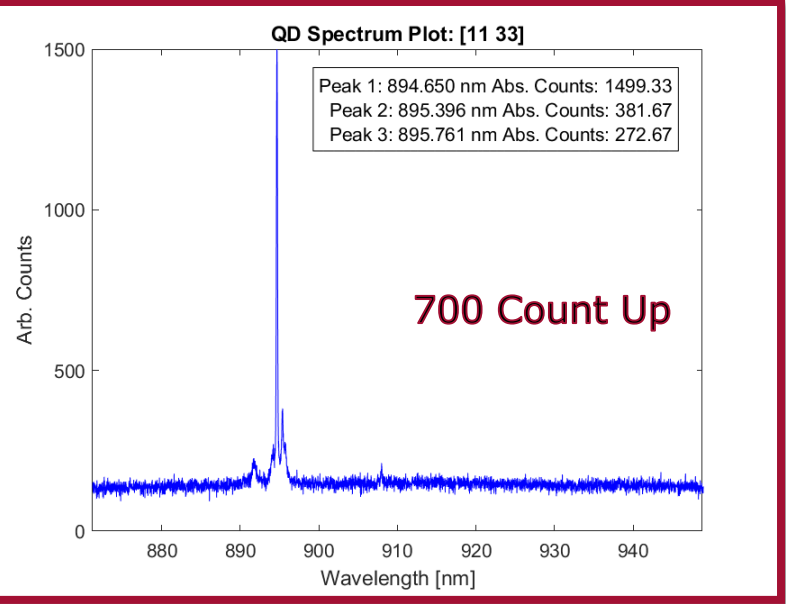
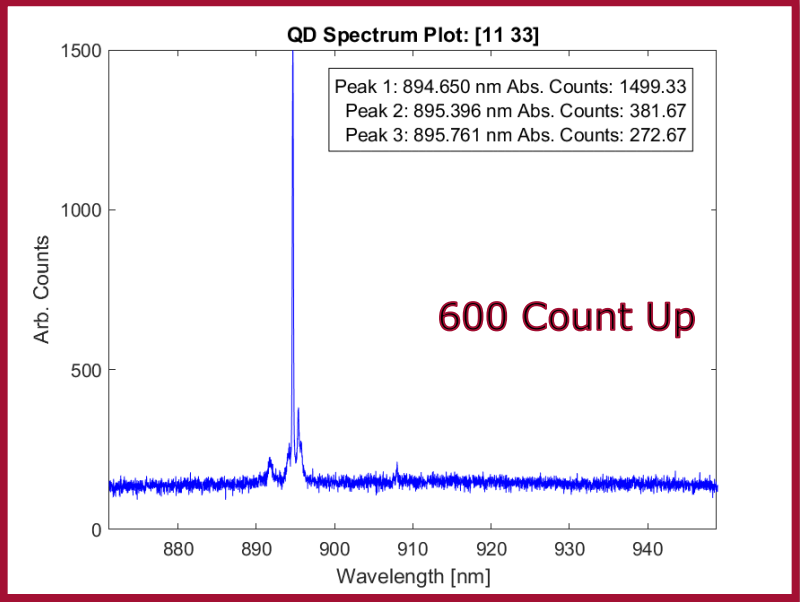
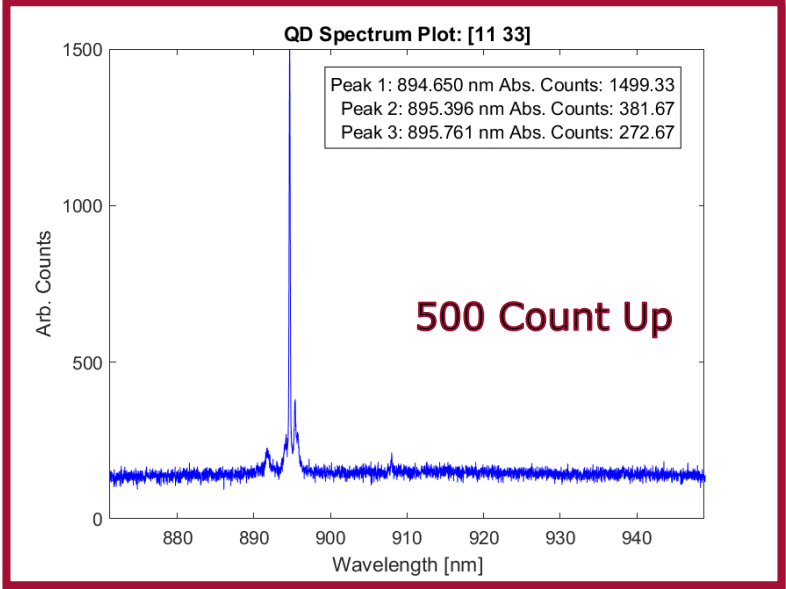
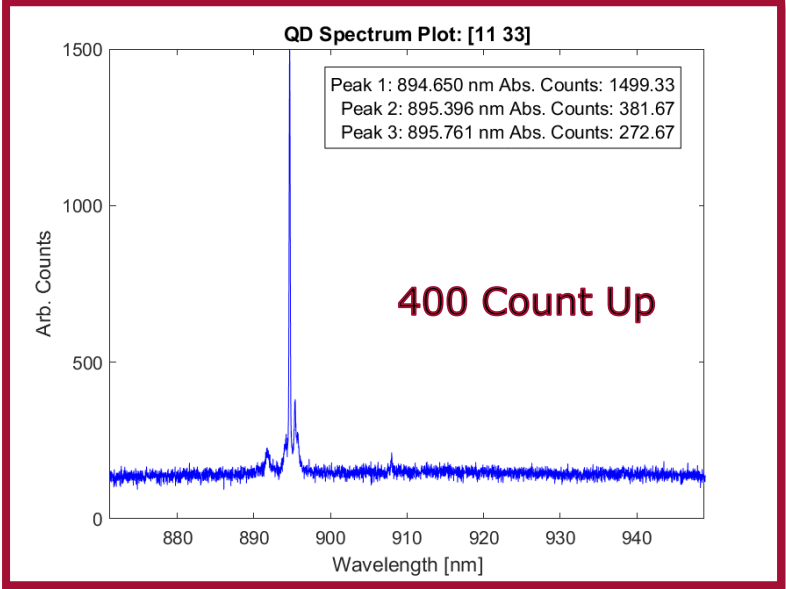


Wavelength (nm)

Data/Results

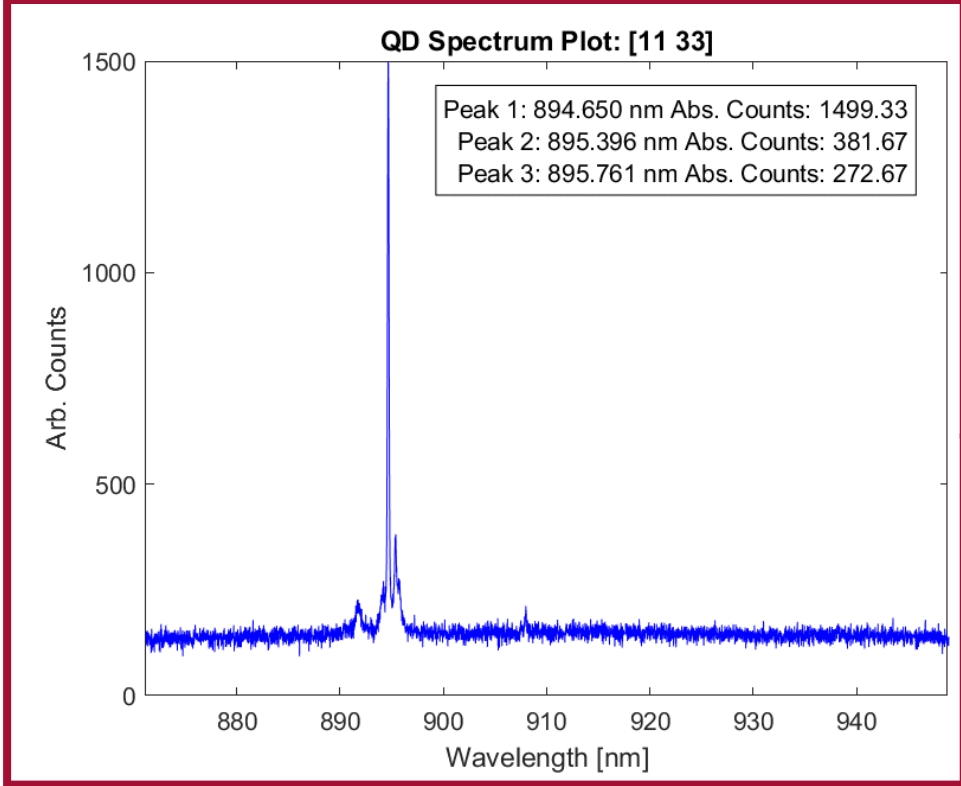


Data/Results

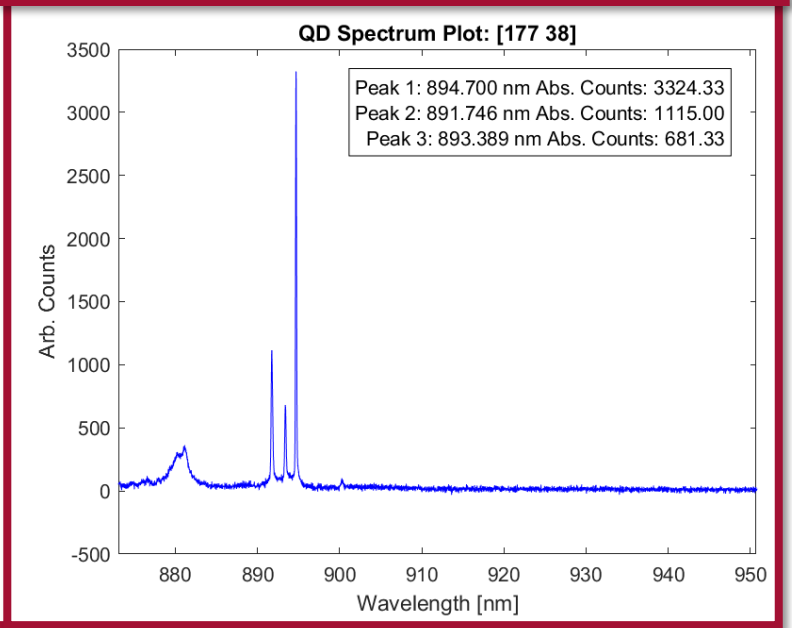
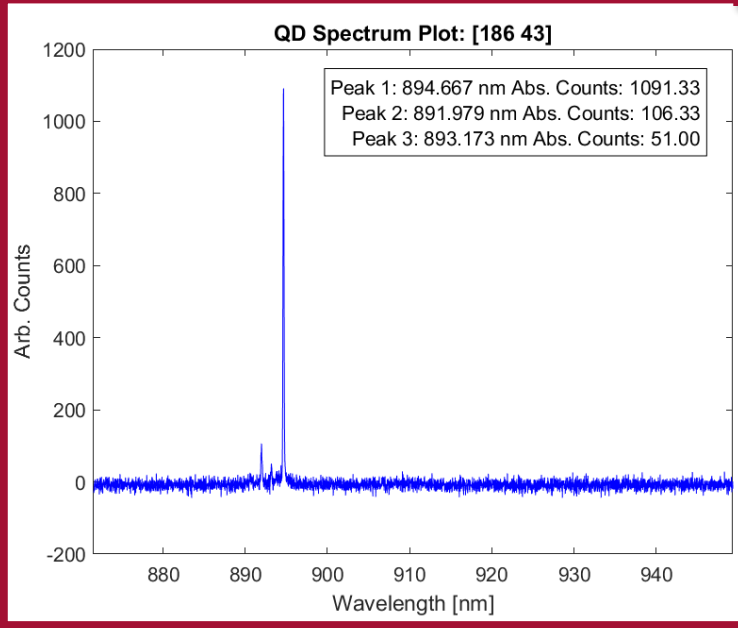
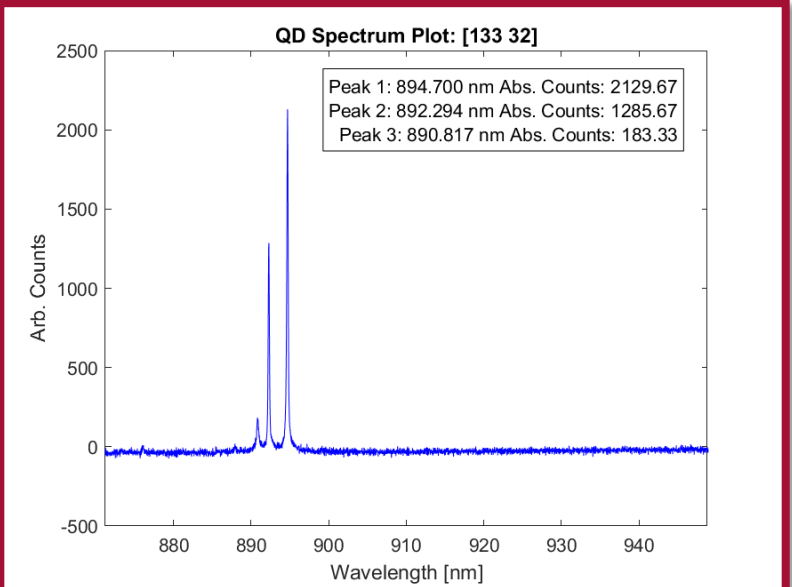
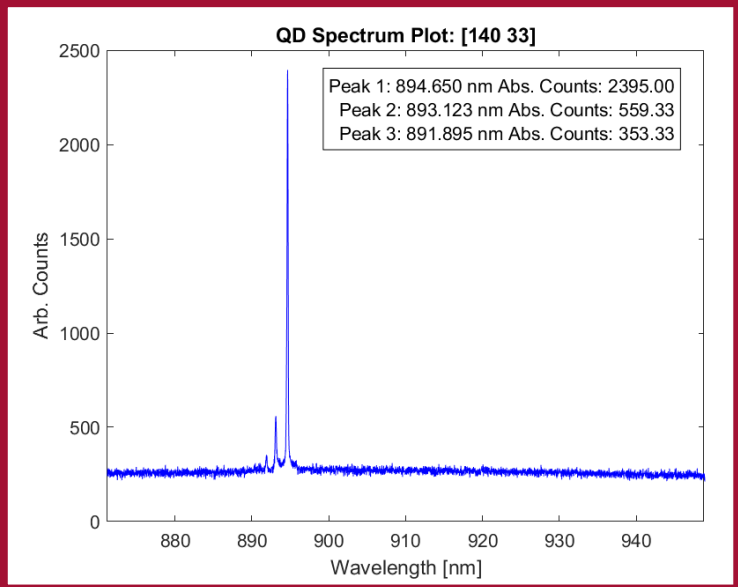


Data/Results

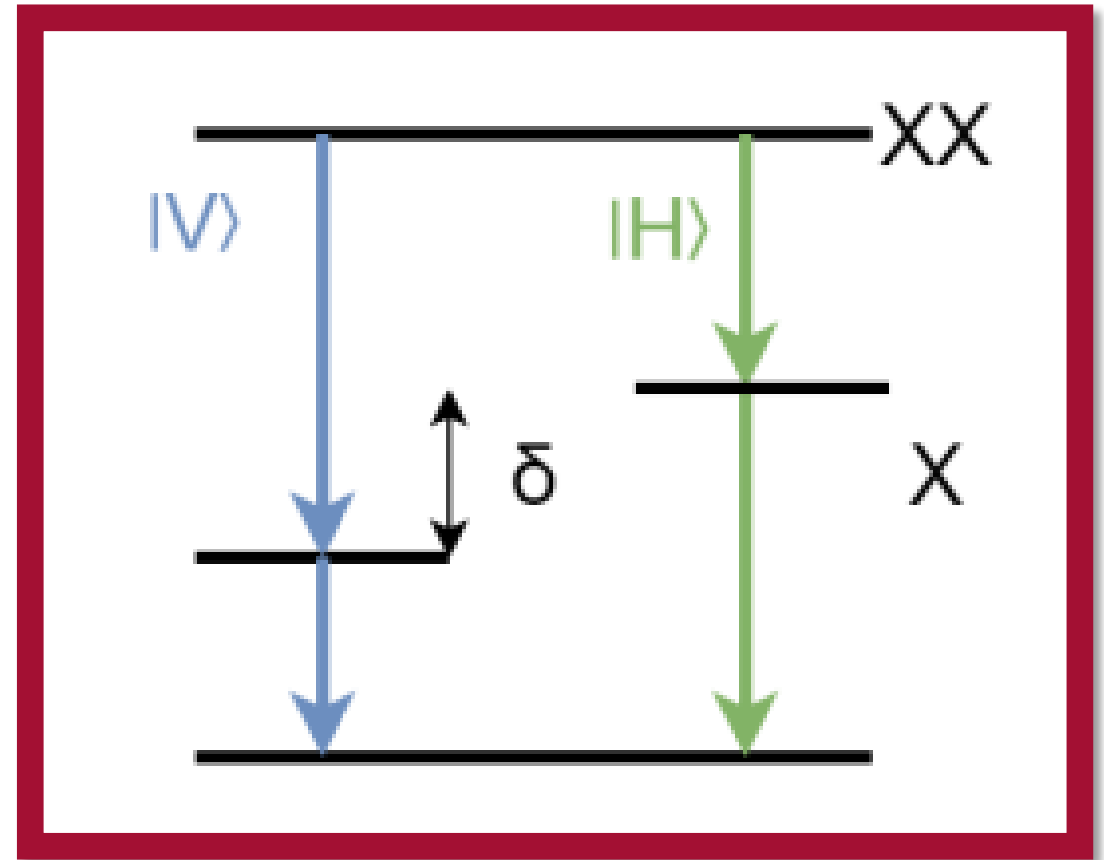
1000 Count Up (Golden Stuff)



Plots that Stood Out



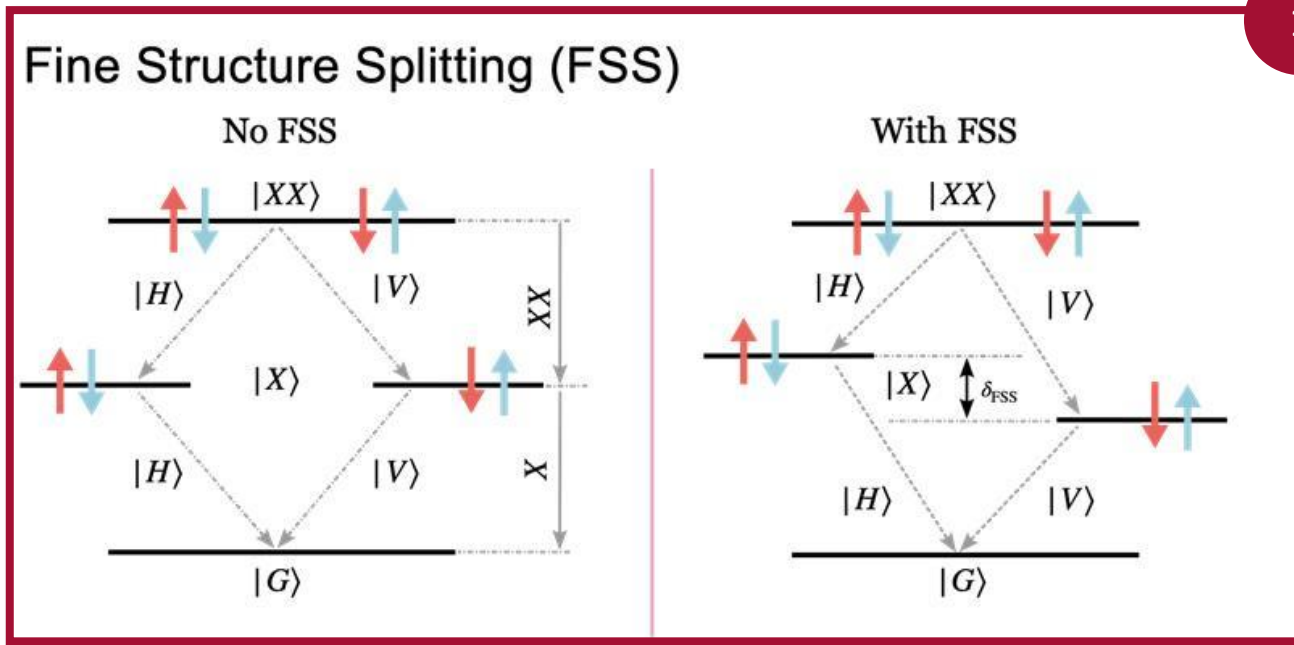
Fine Structure Splitting (FSS) Overview



Where $\delta = \text{FSS Energy}$

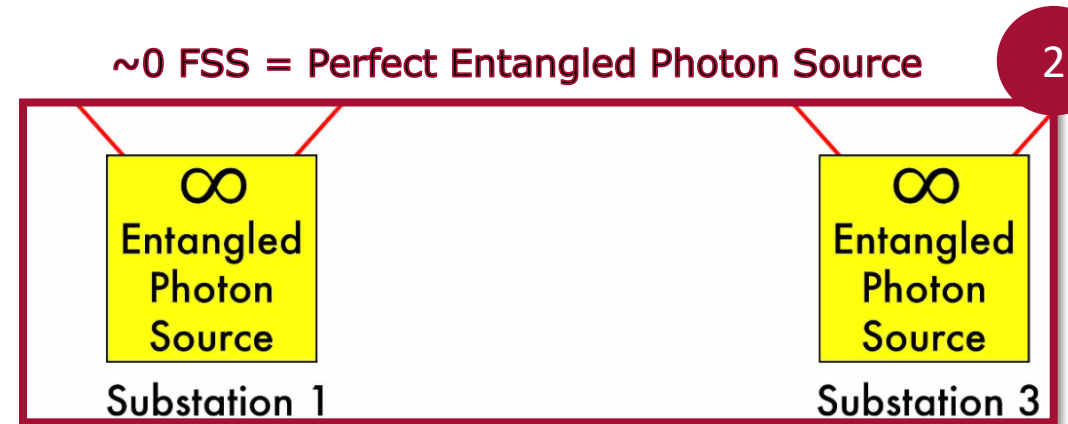
What is FSS?

- Electron-pair recombination leads to **horizontal or vertical polarization**.
- Exciton splits into **two slightly different energy states** depending on the polarization orientation (**FSS energy**)



Why Care About FSS?

- **Entangled photon pairs** generated via **biexciton-exciton cascade**.
- Exciton states must be **degenerate** (same energy no matter polarization orientation)
- Ideal case, finding a **QD with zero FSS**



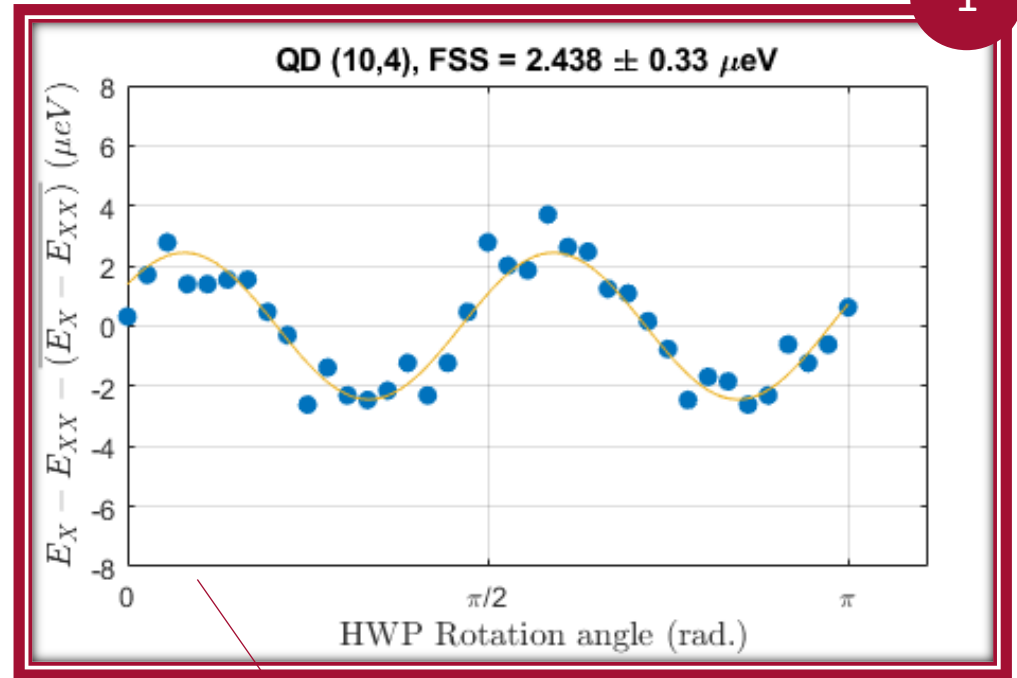
Calculating FSS

FSS Concept: Fine-structure splitting (FSS) changes QD eigenstates from circular to rectilinear.

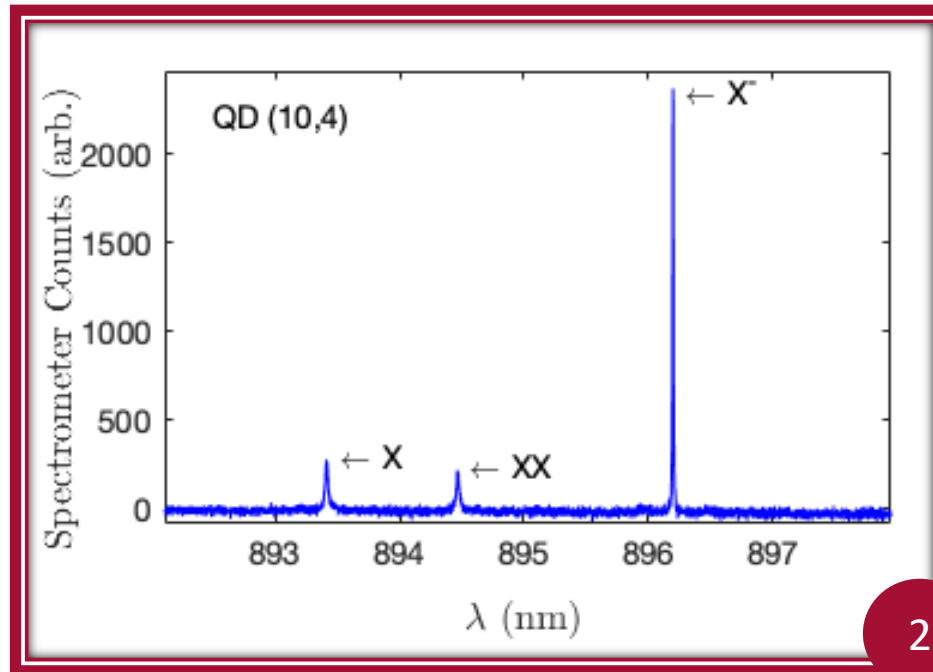
Peak Shift Measurement: Exciton (E_x) and biexciton (E_{xx}) peak shifts recorded for different polarizations.

Data Analysis: ($E_x - E_{xx}$) differences plotted against polarization angle.

FSS Extraction: Amplitude of a fitted cosine function gives the FSS value.



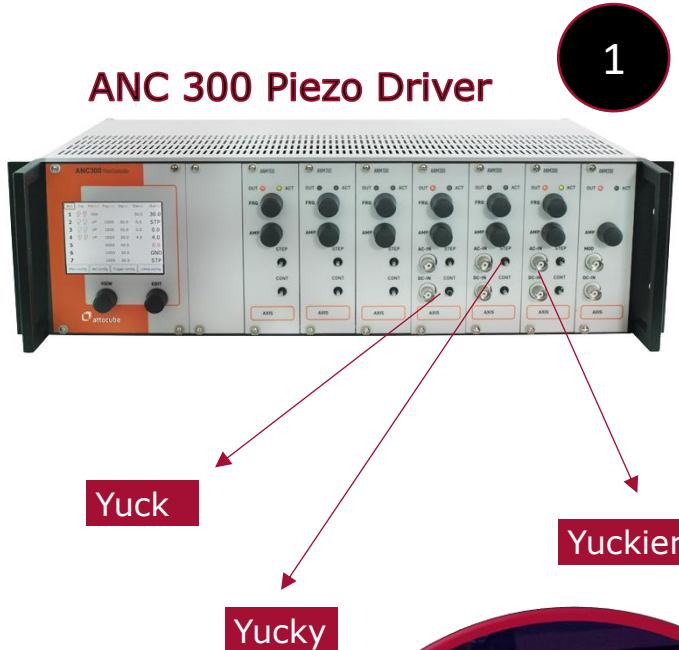
The axis would actually be from 0 to 2π



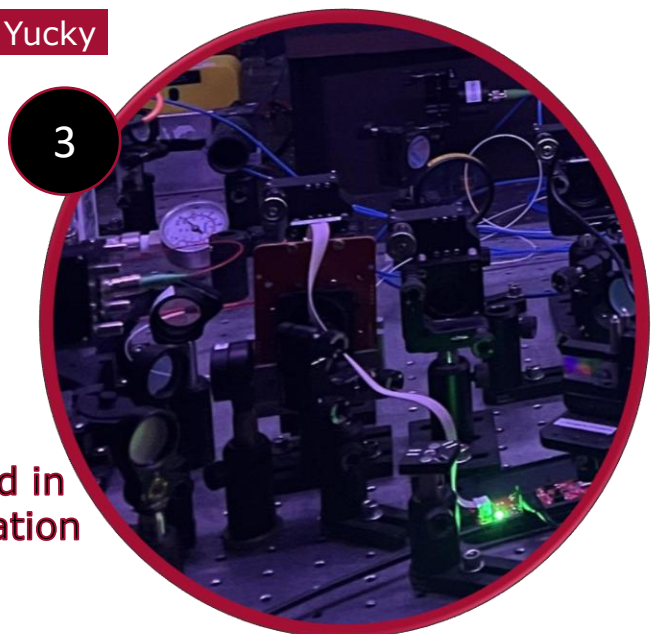
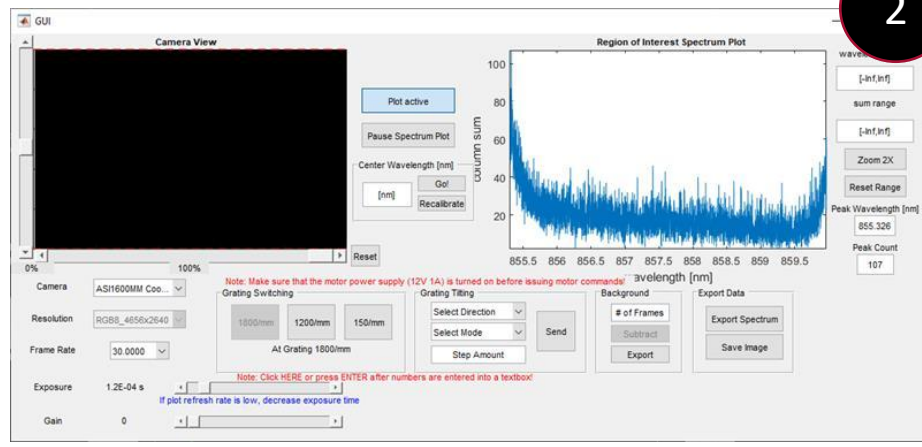
Process to Gather FSS Data

Manual/Old Method

1. Navigate to the quantum dot (QD) of interest using the piezoelectric driver, ensuring precise but gradual positioning.
2. Utilize the live spectrometry script (*FWHM_ver_Richard.m*) to manually align the QD until the emission peak reaches sufficient intensity.
3. Initiate rotation of the half-wave plate (HWP) to monitor and record the wavelength shifts of the exciton and biexciton peaks.
4. Repeat the procedure for each subsequent QD of interest.



Live Spectrometry Script FWHM_Ver_Richard.m (Designed by Jackson and Richard)



- 4
- [105 16]
 - [11 33]
 - [133 32]
 - [139 21]
 - [140 33]
 - [141 22]
 - [144 35]
 - [150 35]
 - [154 16]
 - [154 22]
 - [167 38]
 - [169 38]
 - [170 34]
- ...

Long Tedious list of QDs of Interest

Process to Gather FSS Data

Automated/Improved Method

1. Select **QD manually via row/column input**
or upload a list via text file using the **Upload List** button
2. Click **Show List** to view and confirm selected QDs
3. In the **Process** section:
Select **FSS**
Choose desired total angle: **90°, 180°, 270°, or 360°**
4. Click **Start/Stop Movement** to begin the process and watch it work its magic

UI Tab for Controlling FSS

Raster Scan Specific Movement LED Find Manual Updating Camera Settings List QD ANC300 +

List of QD

[] Save Photon Count Scan Figure ☐ Upload List
ROW COLUMN Skip Brightest Spot Around QD Finding ☐ Empty List
Add To List Remove Last QD Save Raw Img Yes FSS
Show List Start/Stop Movement FSS Total Angle 90

Update Text Will Appear Here

Current QD Plot

Creating FSS Plots

- Both **Manual and Automated FSS methods** require data processing to achieve the **desired FSS plots**
- Requires an important component; **exciton and biexciton identification**

How to Identify Exciton and Biexciton

Exciton (X) and Charged Exciton (X^+ / X^-)

- Both Emission intensity follows a **quadratic** dependence on power ($I \propto P^2$). (Linear at low power)
- X** is differentiated from X^- as usually X^- is more redshifted (lower energy)

Biexciton (XX)

- Emission intensity increases **linearly** with excitation power ($I \propto P^1$).

Legend

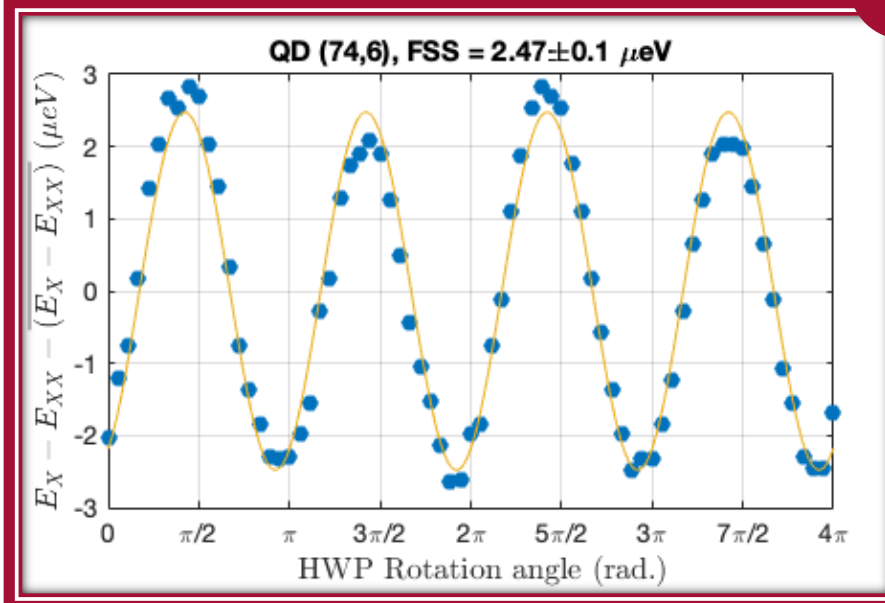
Exciton ■

Biexciton ■

Charged Exciton ■

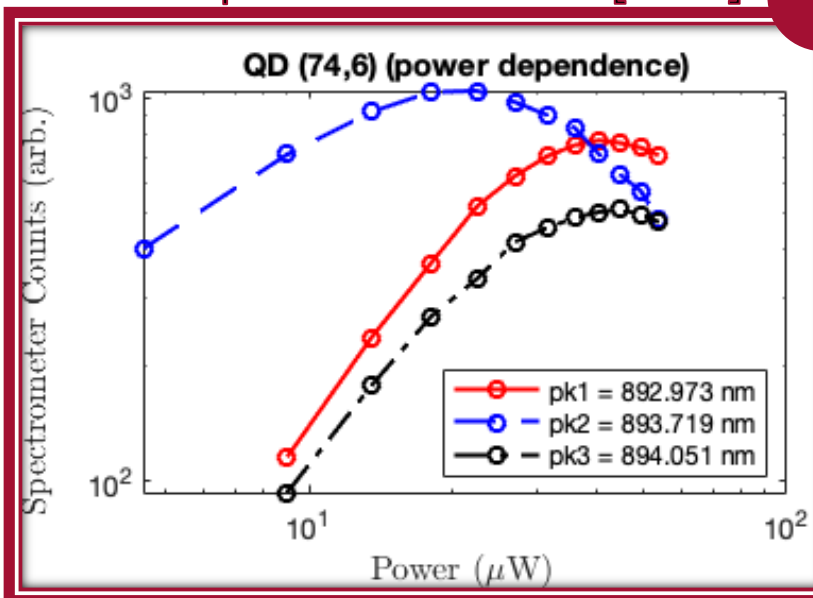
FSS Plot for [74 6]

1



Power Dependence Plot for [74 6]

2



Data/Results

- Identified and compiled a **list of 23 quantum dots** (QDs)
 - Selection based on **proximity to cesium D1 transition wavelength**
 - Prioritized QDs with **high photon count rates**
- Noted **discrepancies compared to 150 nm grating raster scan**
- 13 Spectra scans completed
- 4 full FSS measurements completed on the following QDs:

Wavelength Difference from closest Energy Level

- **[35 36]**
- **[74 46]**
- **[175 24]**
- **[194 27]**

```
[[1 31]] | Peak: 894.583 nm | Count: 483.33 | Δ: 0.005 nm | Completion Status: Spectra Done (Peak at 894.884nm, bad)
[[7 20]] | Peak: 894.583 nm | Count: 1307.33 | Δ: 0.005 nm | Completion Status: Spectra Done (Peak at 894.905nm, bad)
[[74 46]] | Peak: 894.583 nm | Count: 2892.00 | Δ: 0.005 nm | Completion Status: Spectra and FSS Done (overlap for both)
[[111 22]] | Peak: 894.567 nm | Count: 498.00 | Δ: 0.011 nm | Completion Status: Spectra Done (weak peak, 894.713nm)
[[175 24]] | Peak: 894.567 nm | Count: 1643.67 | Δ: 0.011 nm | Completion Status: Spectra and FSS Done (good peak, 894.733nm)
[[45 16]] | Peak: 894.567 nm | Count: 489.00 | Δ: 0.011 nm | Completion Status:
[[71 39]] | Peak: 894.567 nm | Count: 843.33 | Δ: 0.011 nm | Completion Status:
[[154 22]] | Peak: 894.600 nm | Count: 1143.67 | Δ: 0.022 nm | Completion Status: (overlap for both)
[[194 27]] | Peak: 894.600 nm | Count: 1054.00 | Δ: 0.022 nm | Completion Status: Spectra and FSS Done (overlap for both)
[[103 28]] | Peak: 894.550 nm | Count: 927.00 | Δ: 0.028 nm | Completion Status:
[[170 34]] | Peak: 894.617 nm | Count: 3540.33 | Δ: 0.012 nm | Completion Status: Spectra Done
[[38 24]] | Peak: 894.617 nm | Count: 683.67 | Δ: 0.012 nm | Completion Status:
[[57 32]] | Peak: 894.617 nm | Count: 632.67 | Δ: 0.012 nm | Completion Status: (really good)
[[35 36]] | Peak: 894.617 nm | Count: 2581.67 | Δ: 0.012 nm | Completion Status: Spectra and FSS Done (good peak, 894.748nm)
[[1 31]] | Peak: 894.583 nm | Count: 483.33 | Δ: 0.022 nm | Completion Status:
[[7 20]] | Peak: 894.583 nm | Count: 1307.33 | Δ: 0.022 nm | Completion Status: (really good)
[[150 35]] | Peak: 894.633 nm | Count: 2730.33 | Δ: 0.028 nm | Completion Status: Spectra Done
[[28 19]] | Peak: 894.633 nm | Count: 1820.33 | Δ: 0.028 nm | Completion Status:
[[55 23]] | Peak: 894.633 nm | Count: 679.33 | Δ: 0.028 nm | Completion Status:
[133 32]: Spectra Done
[140 33]: Spectra Done
[177 38]: Spectra Done
[186 43]: Spectra Done
```

Discrepancy Filled Plots

Part 1

1800 lines/mm grating (Raster Scan)

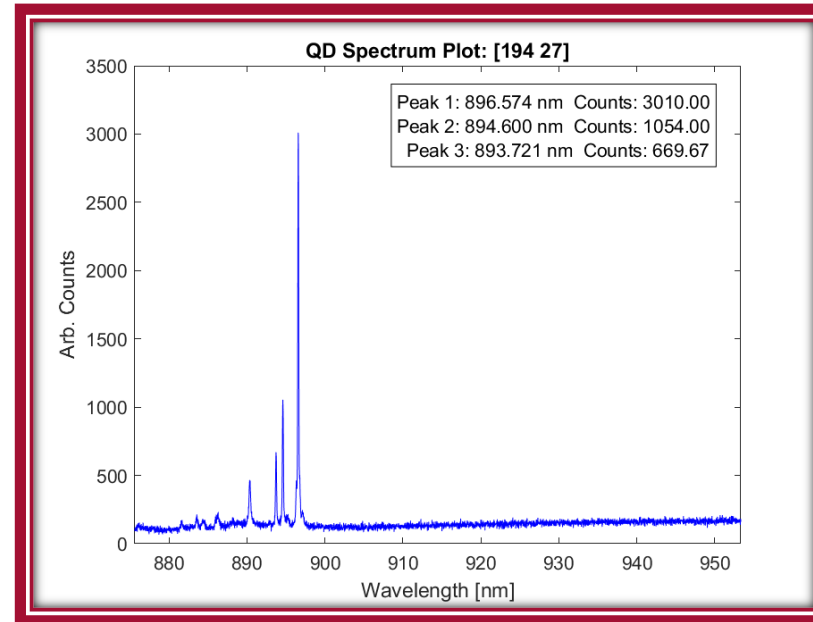
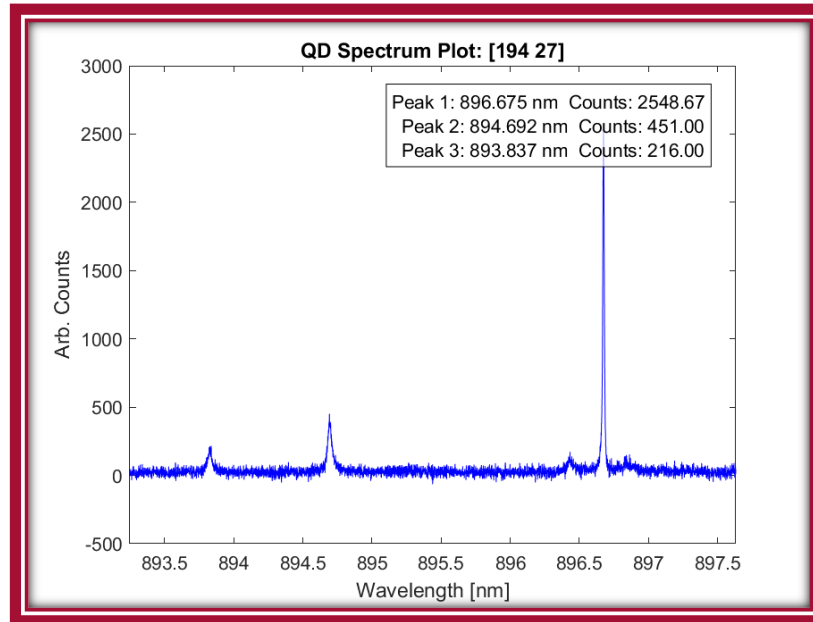
150 lines/mm grating

Wavelength
Difference
Between Same
Peaks of Interest

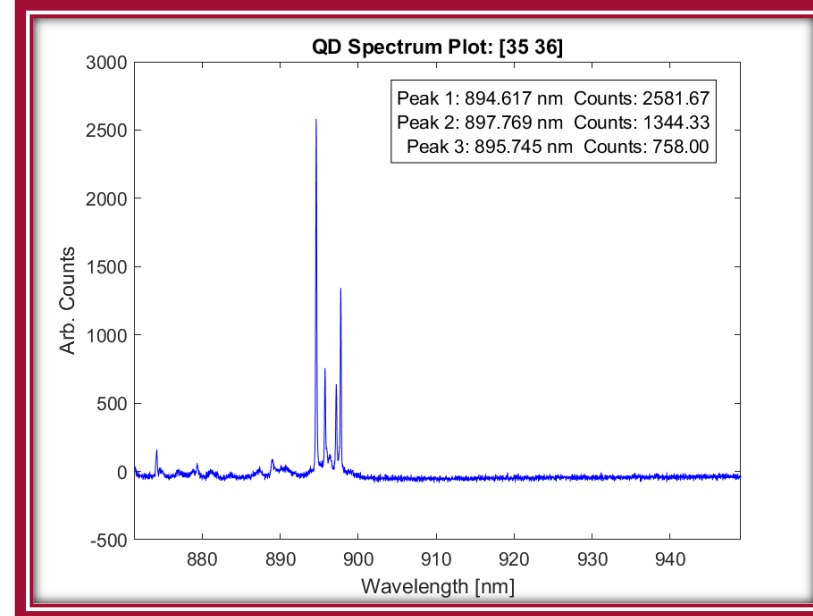
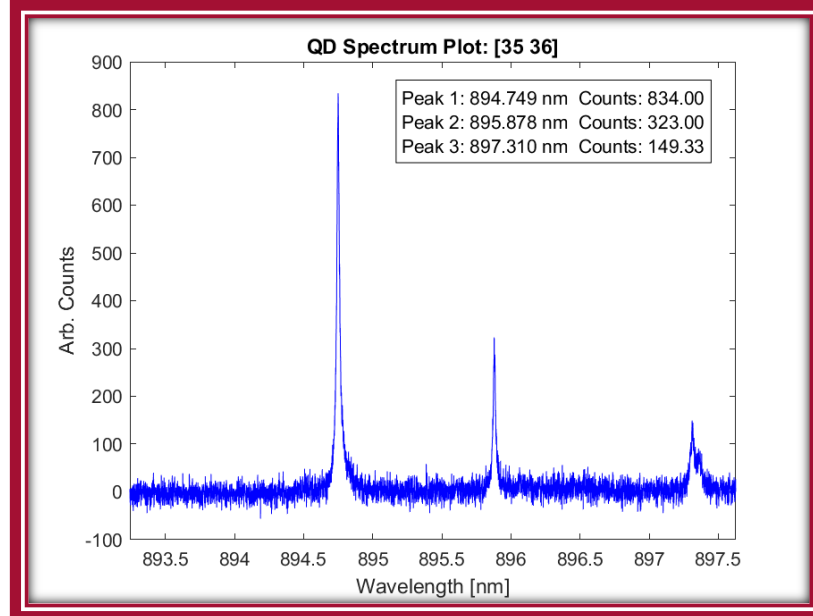
101 pm Peak 1
92 pm Peak 2

132 pm

[194 7]



[35 36]



Discrepancy Filled Plots

part 2

1800 lines/mm grating (Raster Scan)

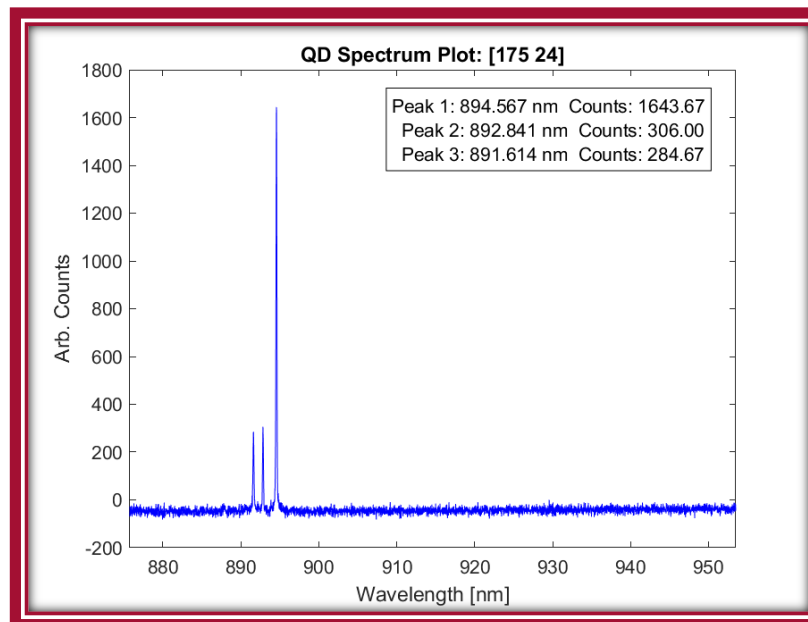
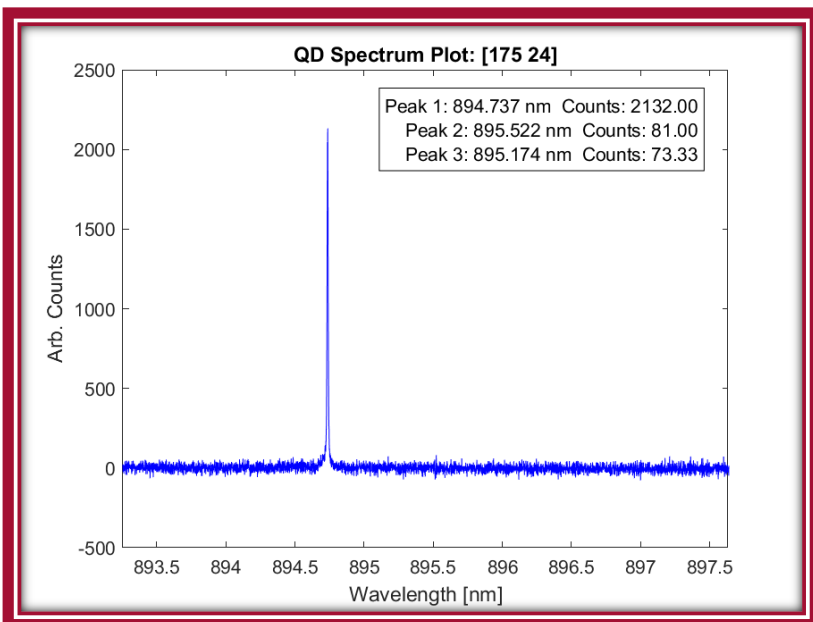
150 lines/mm grating

Wavelength
Difference
Between Same
Peaks of Interest

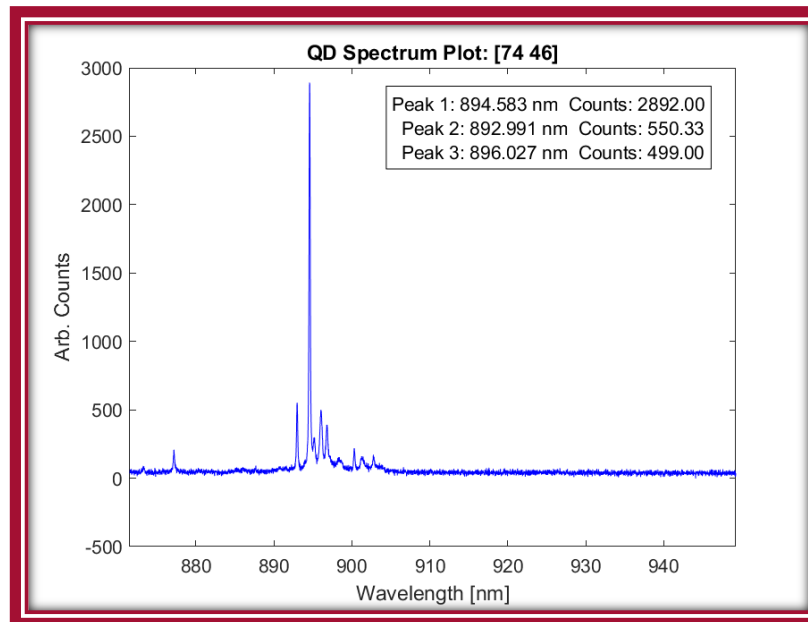
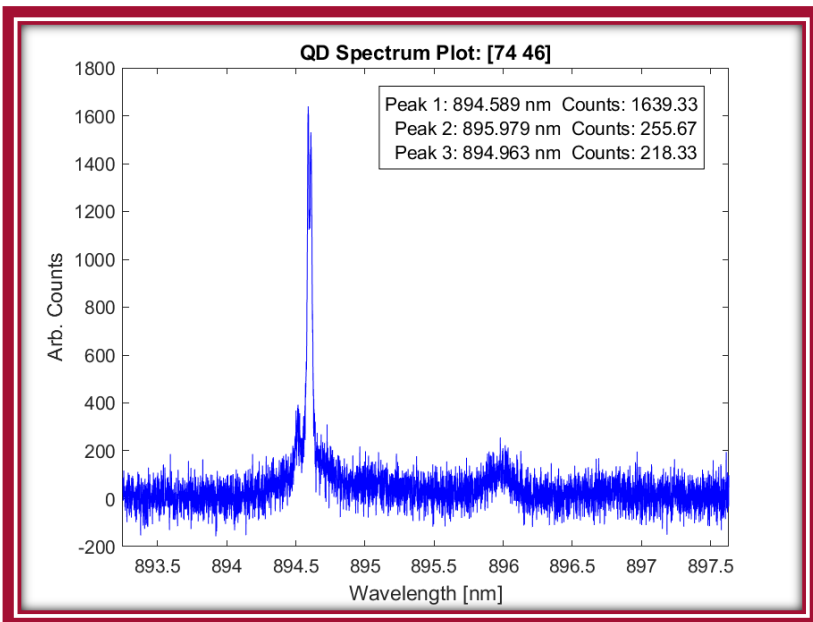
170 pm

6 pm

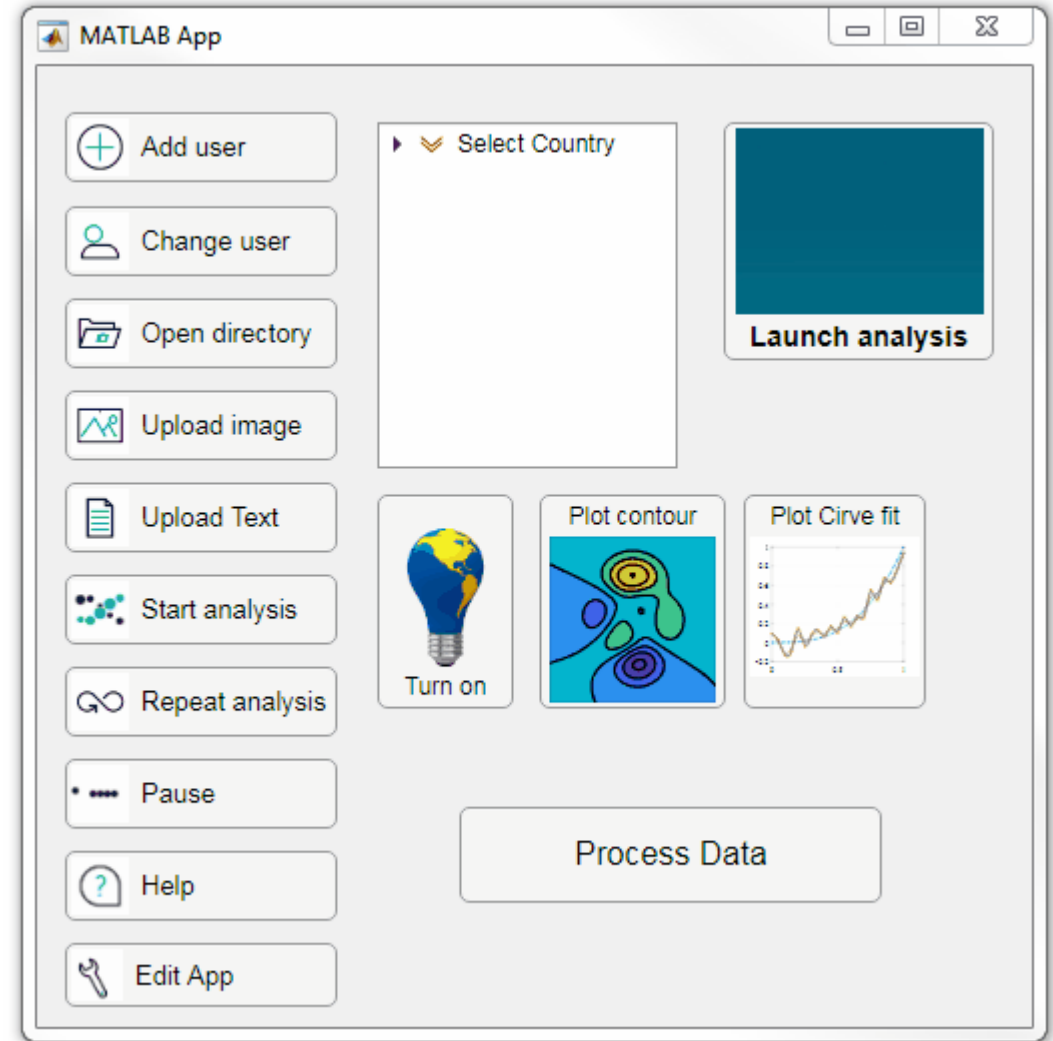
[175 24]



[74 46]



NWQD Operation Application Overview



Overview

- **Actual file name: NWQD_Operating_App.mlapp**
 - Where, **NWQD** stands for **Nano-wire Quantum Dots**
- Sophisticated NWQD application that offer 7 different tabs;

I. Raster Scan

II. Specific Movement

III. LED Find

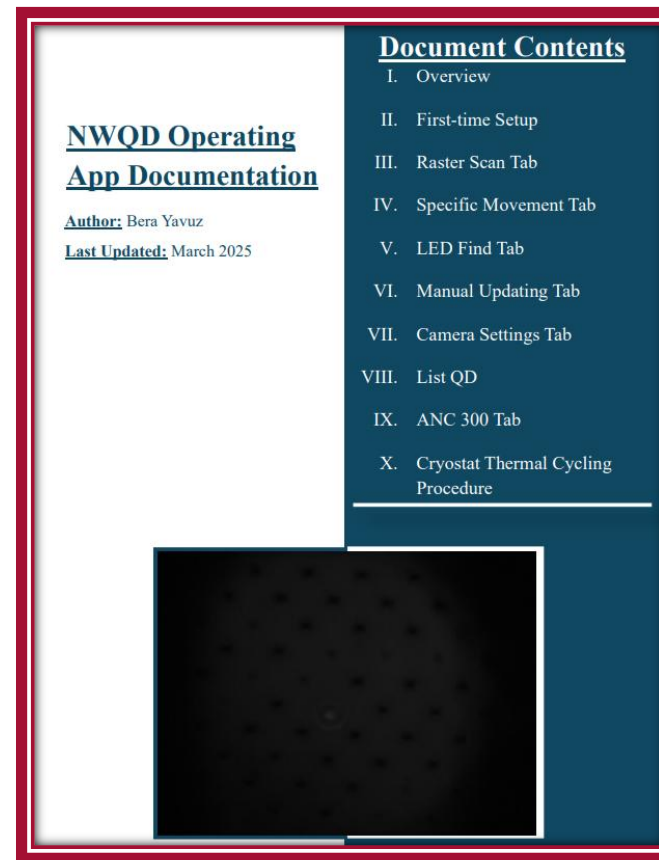
IV. Manual Updating

V. Camera Settings

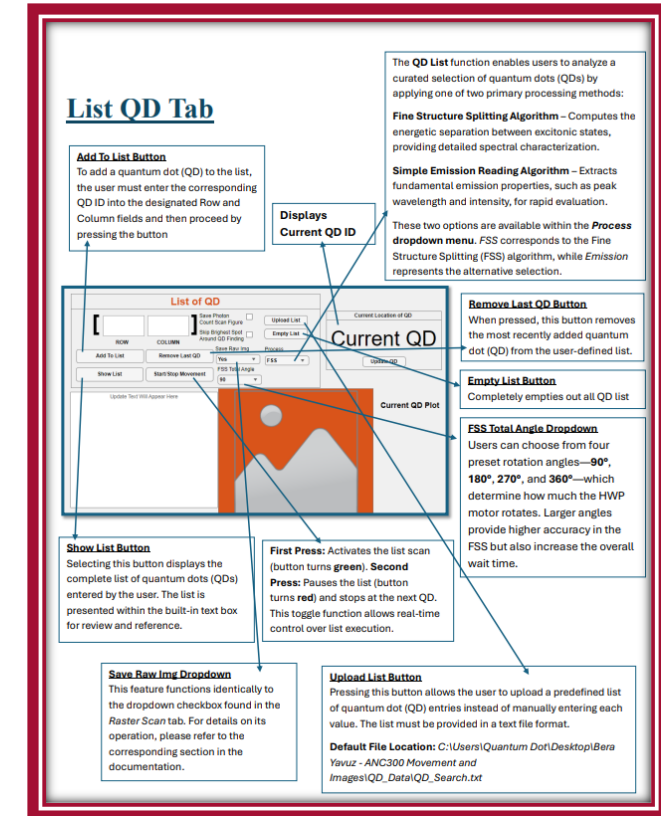
VI. List QD

VII. ANC300

- Saves QD position and selected Region of Interest for Spectrometer Camera Between Sessions
- For more information **Refer to the 12-page Documentation**



Title Page



Example of a Detailed Overview of a function

Raster Scan Tab

- Offers User Friendly approach of **raster scanning a desired QD region**

Can be stopped and started at any time

Updates in Live Time

Raster Scan | Specific Movement | LED Find | Manual Updating | Camera Settings | List QD | ANC300 | +

Raster Scanning Over the NanoWire Chip

Region Selection

Number of Rows

Number of Columns

OK

Start/Stop Raster Scan

☐ Save Raw Images

☐ Complete Current Row

☐ Save Position Images

Current Location of QD

Current QD

Update QD

Update Text Will Appear Here



Current QD Plot

Provides live updates regarding movement, region selected etc.

Specific Movement Tab

Updates in Live Time

- Offers User Friendly approach of **moving to an inquired QD**
- Useful when user is in the **middle of the nanowire chip**, would normally have to go back to [1 1] or other known reference spot

Raster Scan

Specific Movement

LED Find

Manual Updating

Camera Settings

List QD

ANC300

+

Moving to Specific QD on Chip

Inquired QD

[]

ROW

COLUMN

Confirm

No Plot ▼

Start

Current Location of QD

Current QD

Update QD

Specific Movement Settings and Changing (Big Movement)

X Step Number

Confirm Setting

Y Step Number

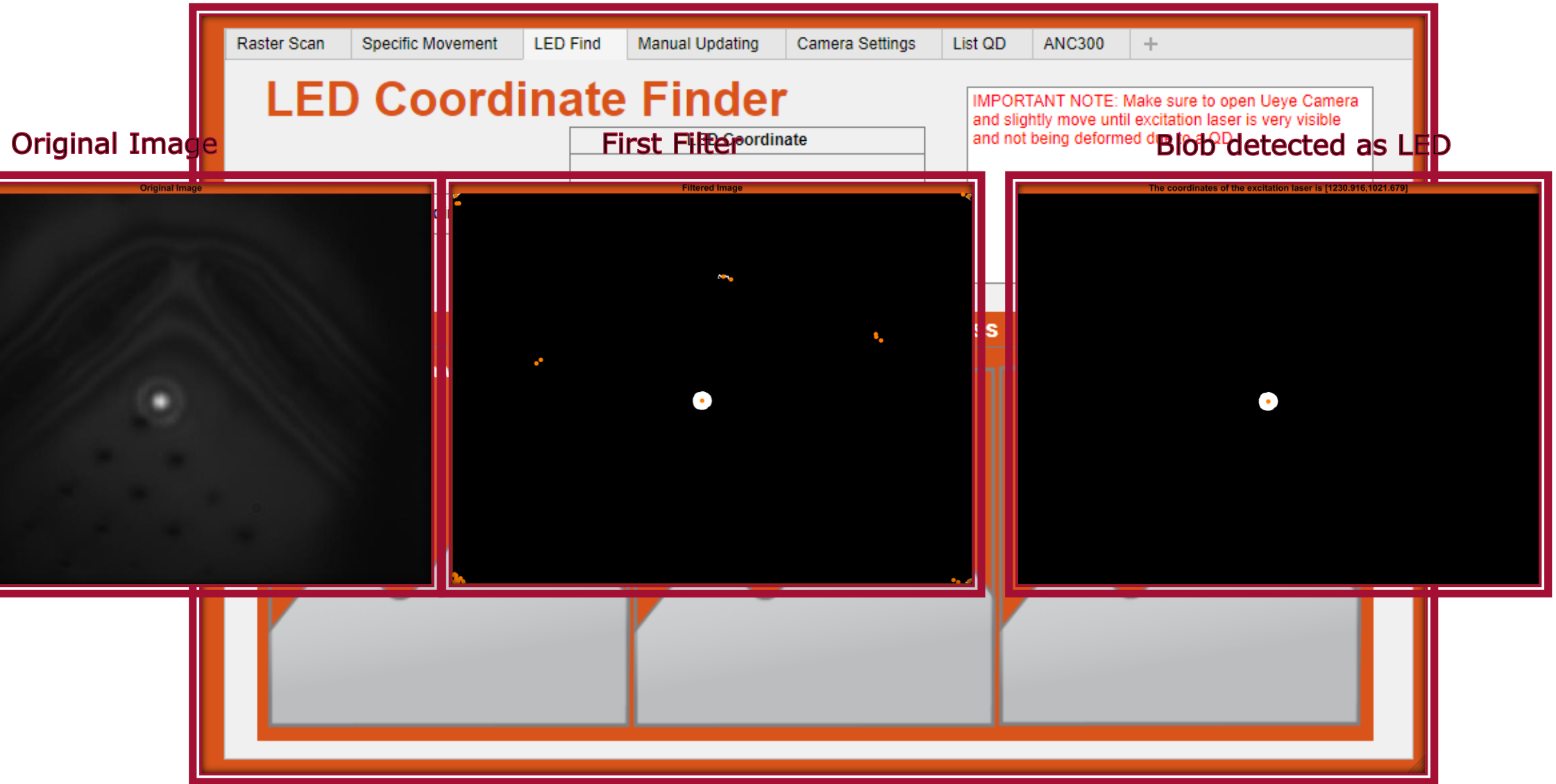
Test X Step Size

Test Y Step Size

Precision Lock

Update Text Will Appear Here

LED Find Tab



Manual Updating Tab

- Manual XY Factor Panel crucial for the **function of whole app**
 - **Needs updating** when **cryostat is cycled**
- Typically, will only need to update **Manual QD panel**
 - Only done if user decides to manually move

The screenshot displays the 'Manual Updating' tab within a software application. The interface features a top navigation bar with tabs: 'Raster Scan', 'Specific Movement', 'LED Find', 'Manual Updating' (selected), 'Camera Settings', 'List QD', 'ANC300', and a '+' icon. The main content area is titled 'Manually Updated Settings' in large orange text. Below this title, there are several interactive panels:

- Manual QD Panel:** Contains two input fields for 'ROW' and 'COLUMN', a dropdown menu for 'Last Direction' (currently set to 'bottomleft'), and a 'Confirm' button.
- Manual XY FACTOR Panel:** Contains two input fields for 'X-Pixels' and 'Y-Pixels', a 'Reverse Move' checkbox, a 'Number of Steps' input field, and a 'Confirm' button.
- Manual LED Coordinate Panel:** Contains two input fields for 'X-Coordinate' and 'Y-Coordinate', and a 'Confirm' button.
- Update Text Will Appear Here:** A large rectangular area for displaying update messages.
- Background Capture for Spectroscopy:** Includes a 'Capture Background Photos' button.
- Capture Current Plot:** A button for capturing the current plot.
- Debug Current Photo:** Includes a 'Capture QD Position Photo' button, a 'Photo Name' input field, and a 'Confirm' button.

Camera Settings Tab

- Recommend to **always have**
Start/Stop UI Camera pressed
 - Displays **live feed of**
nanowire positions

Raster Scan

Specific Movement

LED Find

Manual Updating

Camera Settings

List QD

ANC300

+

ASI (Spectrometer Camera) Settings

Exposure (Seconds)

0102030405060708090100

0

Gain

0102030405060708090100

0

Gamma

0102030405060708090100

0

Brightness

0102030405060708090100

0

Start/Stop UI Camera

Spectrometer Grating

1800mm

Start/Stop Live Spectrum

Start/Stop ASI Camera

Percentage to Cover Below and Above

10%

User can select any value between 2 and 50. Ex. 2 represents 2% of the total screen below and above or 4 percent of the whole screen

Midde Point Line

0

User can select any value between -50 and 50. Ex. 0 represents a line of interest at middle of screen, 50 represents the tippity top and -50 is the bottom of the bar

Configures Placement of Red Lines (Region of interest)

Raw Spectrum Live Feed

List QD Tab

- Allows user to get **FSS data** or **Emission Spectrums** on a list of QDs

Raster ScanSpecific MovementLED FindManual UpdatingCamera SettingsList QDANC300+

List of QD

[]

ROWCOLUMN

Add To List

Remove Last QD

Show List

Start/Stop Movement

Save Photon Count Scan Figure☐

Skip Brightest Spot Around QD Finding☐

Save Raw Img
Yes▼

FSS Total Angle
90▼

Upload List

Empty List

Process
FSS▼

Current Location of QD

Current QD

Update QD

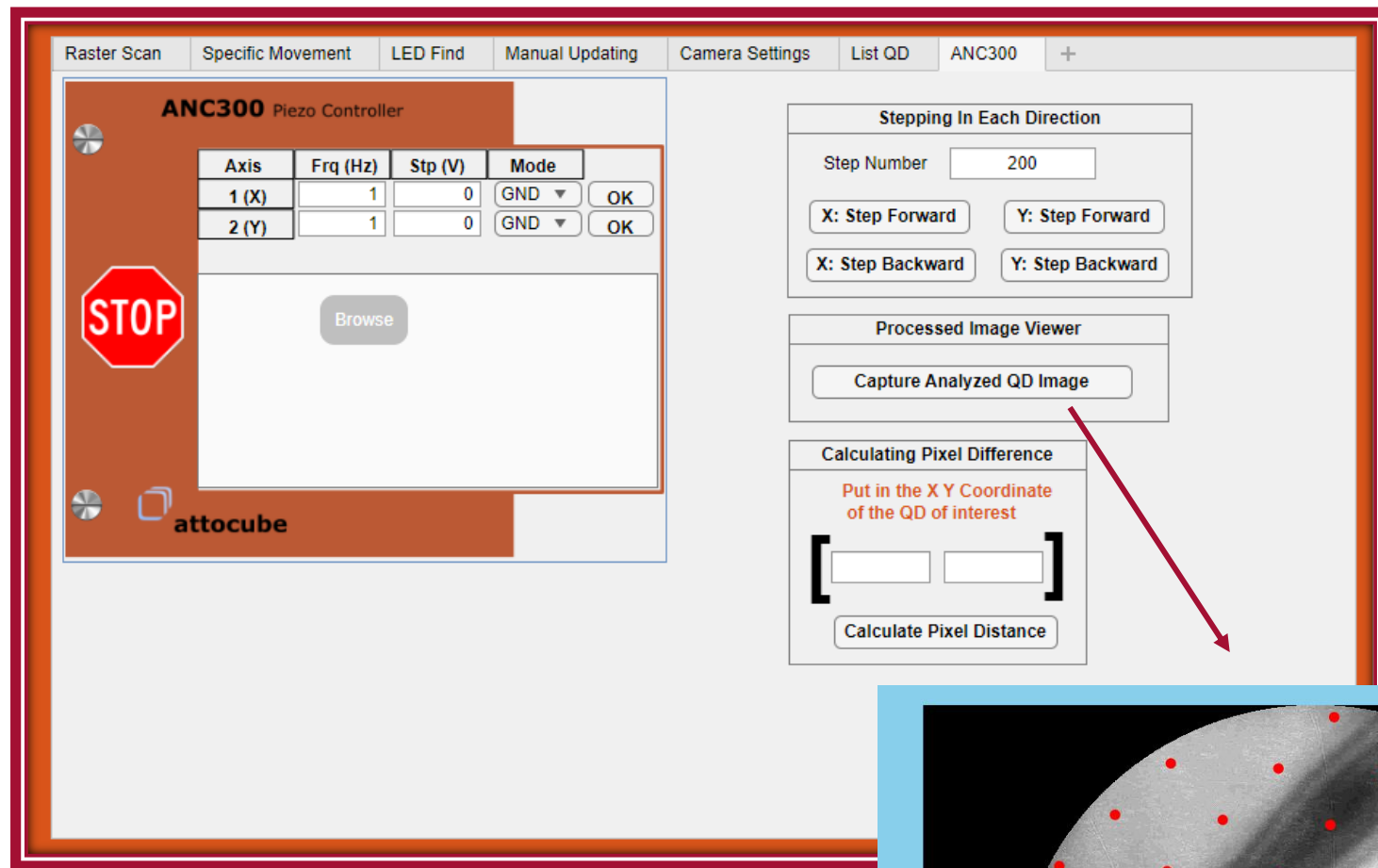
Update Text Will Appear Here



Current QD Plot

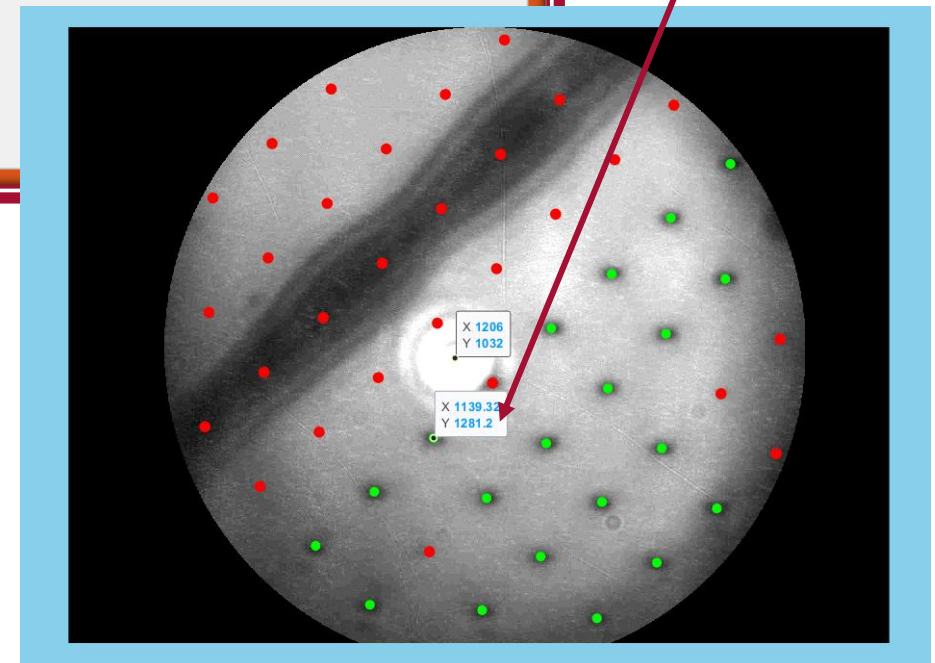
ANC300 Tab

- Crucial for calculating the accurate settings user must put into the **manual updating tab** after **Cryostat cycling**



If this was dot user traveled from, put these coordinates in Calculation Panel

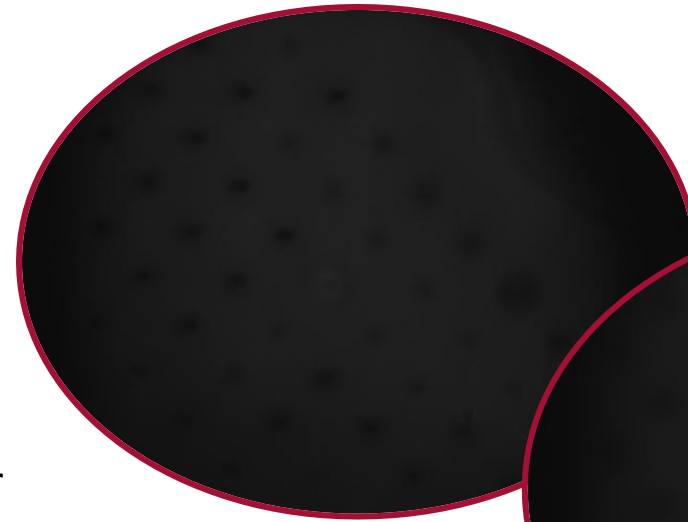
Example of an Analyzed QD Position Image



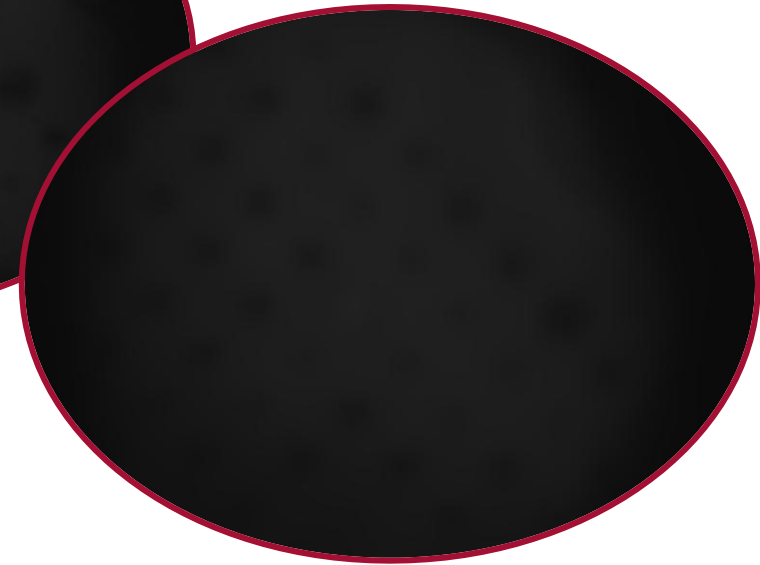
Small Tweaks to Code

- **Issue:** attempt to snap a position photo before movement completed, **blurry image = undetectable dots**
- **Fix at the time:** introduced a manually defined extra pause via testing (**defined as 2.5 seconds**)
- **New Fix:** figured out how to translate byte data from piezo into voltage (**0 = no movement etc.**)
- Saved **time of 2 seconds per dot**
 - translated to **400 seconds saved per column or 7 minutes**
- **Safety Net:** If a movement takes longer than expected — for example, due to an unexpected **power loss or a stuck piezo stage** —algorithm **detects the issue and triggers an emergency stop**

Desired Position Photo



Blurry Position Movement

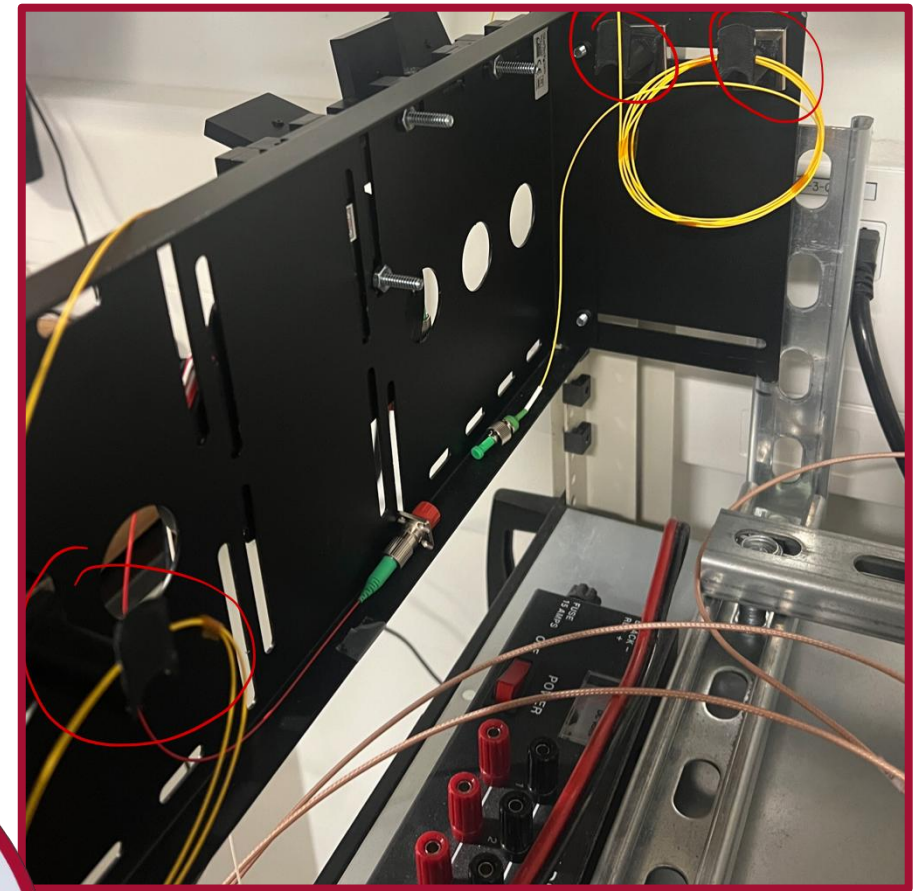
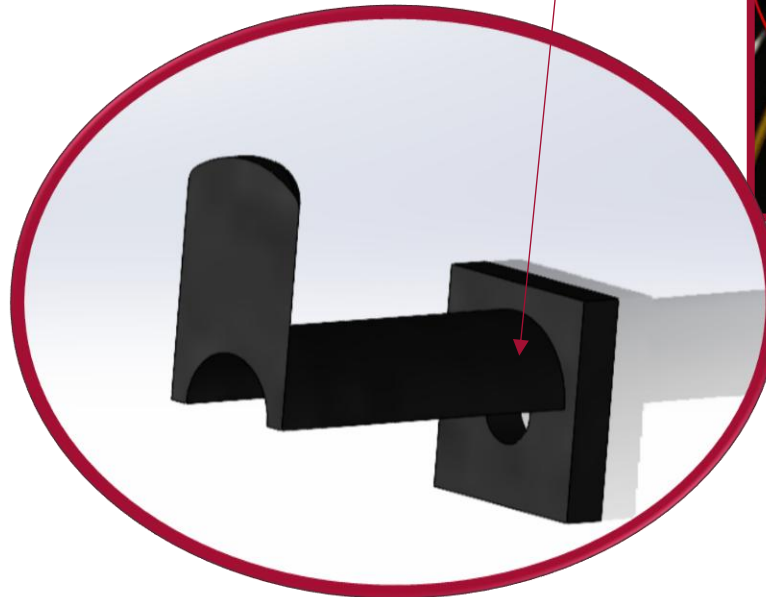
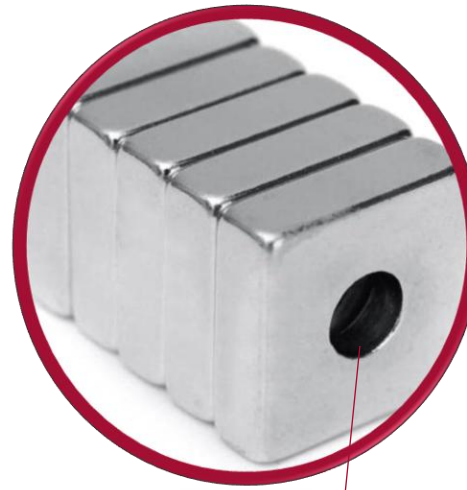


Side Projects



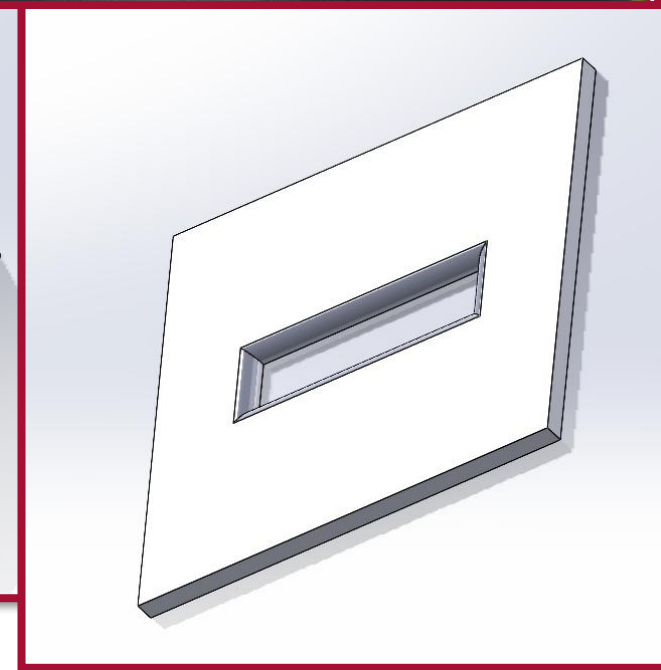
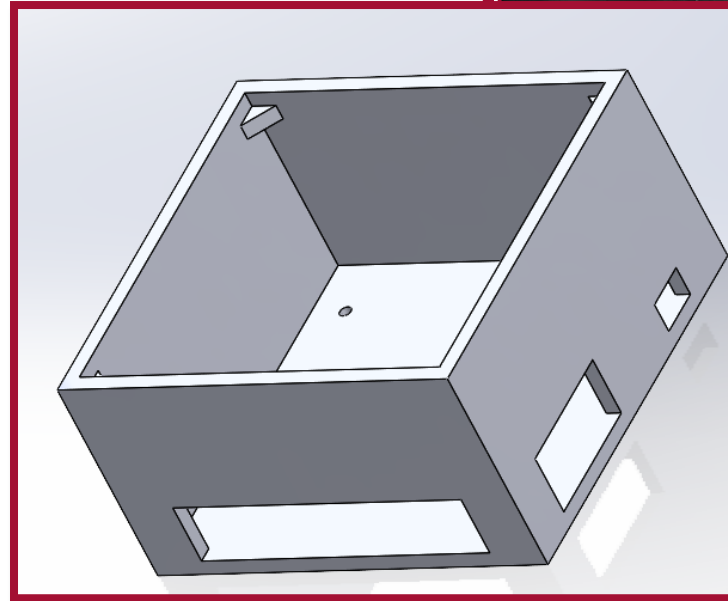
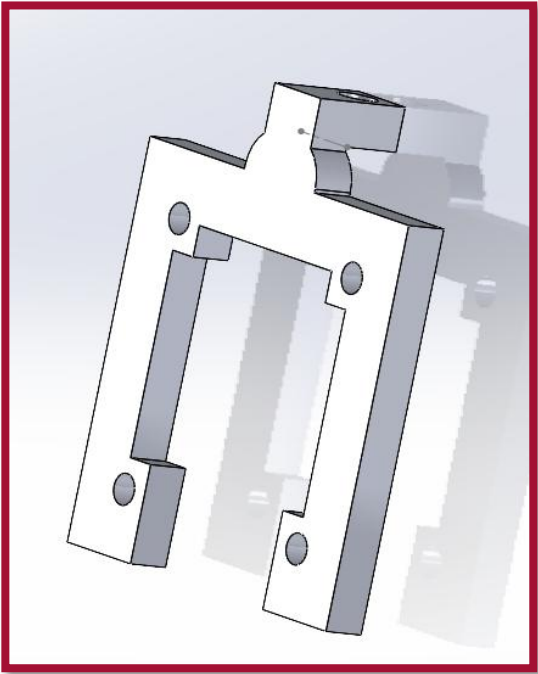
Fiber Optic Organization

- Designed a **handful** of **different Fiber optic** hooks (magnetic and non-magnetic), currently used around the lab



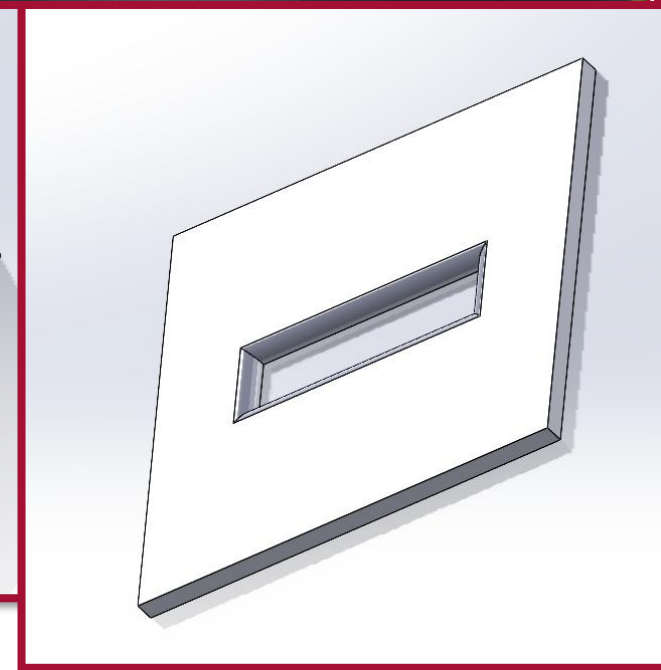
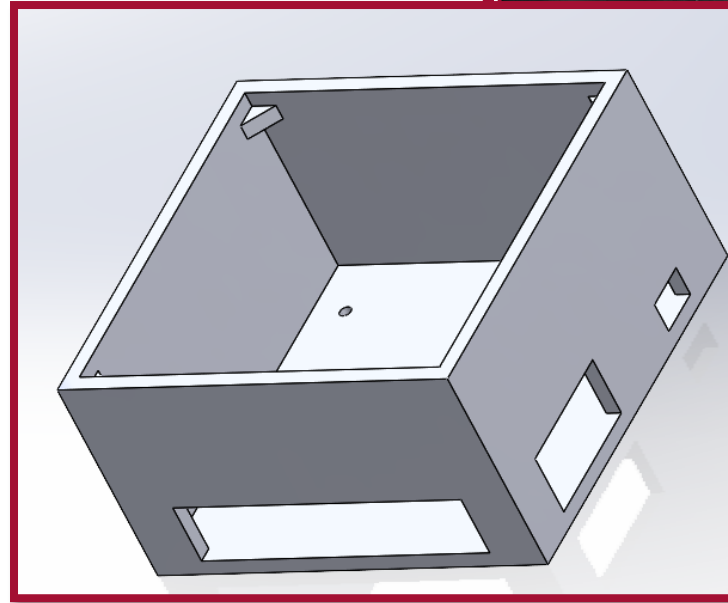
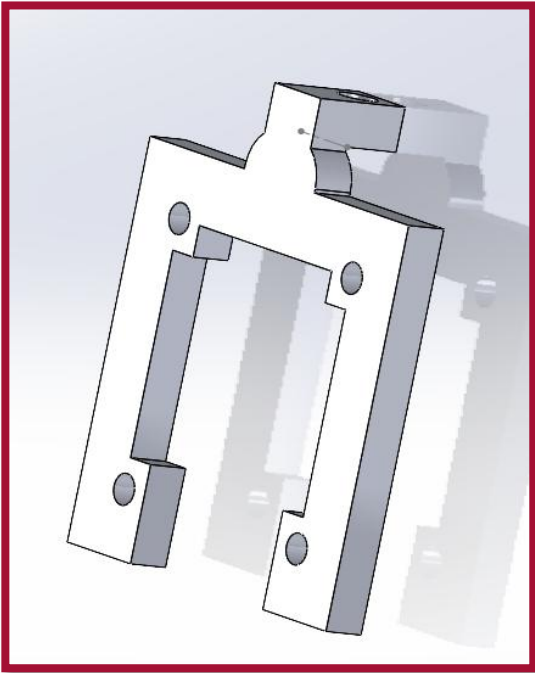
Raspberry Device Holders

- Designed a **simple box** with a **simple lid** for the Raspberry Pi Board that controls Cameras for cold atom setup
- Designed **new mounts** for **Raspberry Pi Cameras**



Raspberry Device Holders

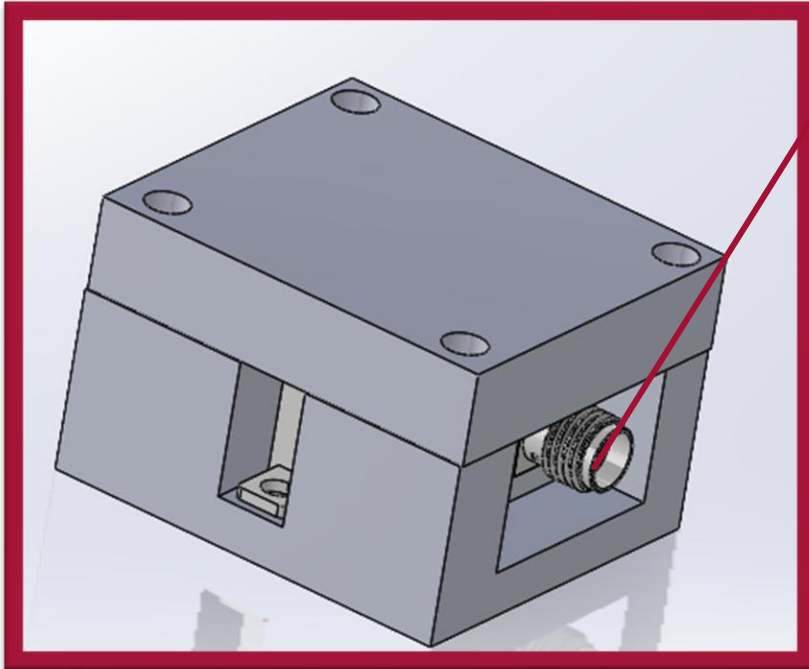
- Designed a **simple box** with a **simple lid** for the Raspberry Pi Board that controls Cameras for cold atom setup
- Designed **new mounts** for **Raspberry Pi Cameras**



Custom New Cheap Photo Detector

- Currently working on designing a new custom high-speed fiber-optic detector
- Already have existing one, however **expensive and thus overkill for laser locking purposes**

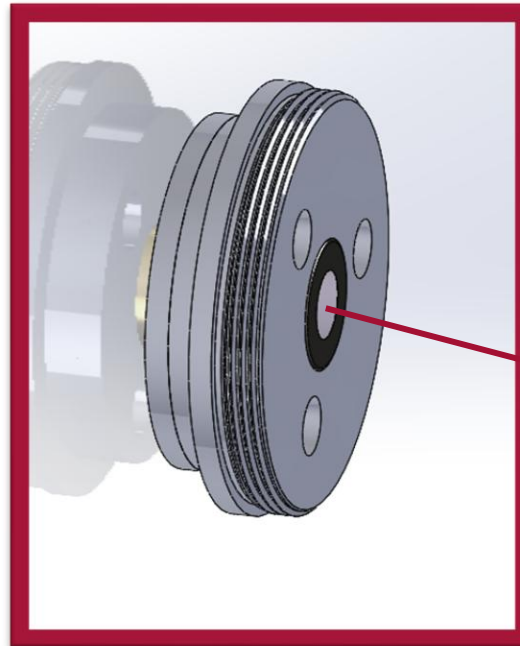
Bias Tee Holder



Bias Tee



Fast Photo Diode Holder



Fast Photo Diode Holder



Photo Detector to Replace

